
LAMP: Data-Efficient Linear Affine Weight-Space Models for Parameter-Controlled 3D Shape Generation and Extrapolation

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Abstract

Generating high-fidelity 3D geometries under explicit parameter constraints is central to engineering design, yet current methods often require large datasets and fail to provide reliable control beyond the training distribution. We introduce LAMP, a data-efficient framework for controllable and interpretable 3D generation that aligns signed distance function (SDF) decoders by overfitting each exemplar from a shared initialization, then generates new designs by solving a parameter-constrained affine mixing problem in the aligned weight space. To improve reliability, we propose a linearity-mismatch safety metric that detects when mixed decoders leave the valid local regime. We evaluate LAMP on DrivAerNet++, Blended-Net, and additional industry-level vehicle families, including sports cars, SUVs, and convertibles. LAMP enables controlled interpolation with as few as 50 samples, safe extrapolation up to 100% beyond training ranges, and performance-guided optimization under fixed parameters, outperforming conditional autoencoder and Deep Network Interpolation (DNI) baselines in extrapolation, data efficiency, and parameter fidelity. Our results demonstrate that LAMP advances controllable, data-efficient, and safe 3D generation for design exploration, dataset generation, and performance-driven optimization.

1. Introduction

Engineering design often requires generating 3D geometries that satisfy explicit, human-interpretable parameters such as

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ramp angle, diffuser angle, roof height, or aerodynamic drag. This setting differs from standard 3D generation: designers typically have few annotated exemplars, need controlled edits beyond the observed range, and must evaluate candidates with expensive downstream tools (Regenwetter et al., 2022). In automotive design, for example, engineers iteratively vary shape parameters and run CFD simulations, which can require hundreds of CPU hours per design in DrivAerNet++ (Elrefaie et al., 2024). Thus, the key bottleneck is not only generation quality, but rapid, controllable, and valid geometry exploration under scarce data.

Existing 3D generative models struggle in this regime (Nichol et al., 2022; Jun & Nichol, 2023; Poole et al., 2022). Many methods learn a shared latent space and seek semantic directions through conditioning, disentanglement, or traversal (Vahdat et al., 2022; Zhao et al., 2023; Morita et al., 2024; Xiang et al., 2025). While effective with large datasets, these models often require thousands to millions of shapes, provide limited guarantees of parameter control, and extrapolate poorly beyond the training distribution. This is especially problematic for engineering pipelines, where the targets are physical parameters and evaluation is expensive.

We introduce **LAMP** (Linear Affine Mixing of Parametric shapes), a data-efficient framework for parameter-controlled 3D mesh generation. LAMP overfits one signed distance function (SDF) decoder per exemplar, starting all decoders from a shared initialization to produce an aligned weight space. Given a target parameter vector, LAMP solves a lightweight affine mixing problem, combines exemplar weights, and decodes the resulting SDF into a mesh. This turns controllable generation into a direct parameter-space solve followed by weight-space mixing.

LAMP is motivated by two observations: (i) many engineering shape parameters act approximately linearly on geometric control structures, and (ii) neural SDF decoders are locally linear in weights near a shared initialization. Together, these allow LAMP to replace heavy conditional generator training with a small linear solve. In practice, this makes the geometry-generation step fast: parameter solving takes under 10 ms and meshing takes seconds. LAMP does not replace CFD or engineering validation; instead, it accelerates the candidate-generation stage, enabling rapid

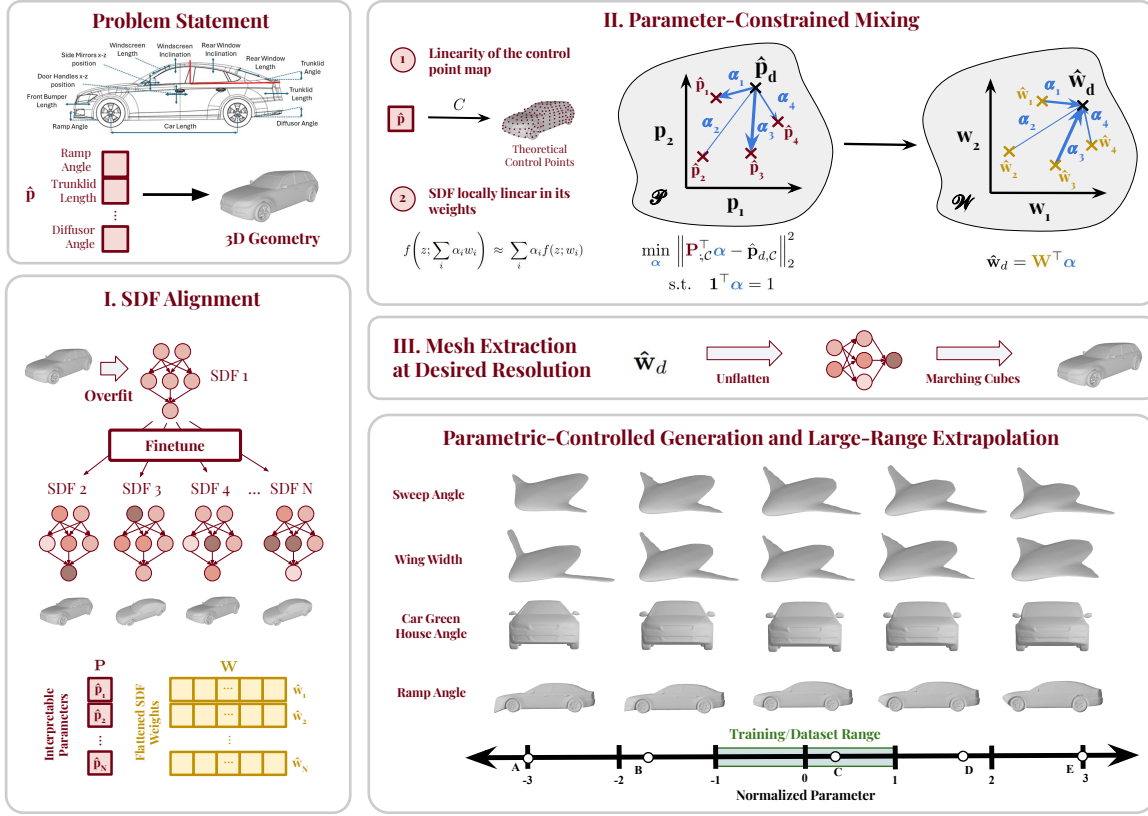


Figure 1. Overview of LAMP: (I) aligned SDF weight space construction, (II) parameter-constrained mixing, and (III) mesh extraction, enabling parametric control and large-range extrapolation.

exploration before downstream simulation.

To improve reliability, we introduce a *linearity-mismatch* safety metric that checks whether a mixed decoder remains in the local linear regime. For sampled 3D points, we compare the output of the mixed SDF decoder with the affine combination of the individual SDF outputs, accepting generations only when the mismatch is below a threshold. Constraint compliance is evaluated using mesh-based surrogate predictors and, when possible, direct geometric measurements on decoded meshes.

We evaluate LAMP on DrivAerNet++ cars and BlendedNet aircraft across interpolation, extrapolation, and performance-guided optimization. On DrivAerNet++, LAMP enables controlled interpolation, large-range extrapolation up to 100% beyond dataset bounds, and drag-aware optimization under fixed parameter constraints. On BlendedNet, LAMP transfers to aircraft geometries, improving both interpolation and extrapolation fidelity. To further test real-world transfer, we add industry-level vehicle experiments on three distinct car categories: a sports car (Toyota Supra), an SUV (Toyota C-HR), and a convertible (Porsche Carrera GT). Using only 50 samples per category and direct mesh measurements, LAMP achieves strong parameter control across

all three families, showing that the method is not tied to one dataset, topology family, or vehicle style.

Overall, LAMP bridges data-efficient generative modeling and practical engineering design. It converts a small set of annotated exemplars into an aligned SDF weight space for fast, interpretable, and safe parameter-controlled generation. This makes it well suited to low-data, parameter-driven design workflows where candidate evaluation is expensive and controllable extrapolation is essential. Code is available at: <https://github.com/ghadinehme/LAMP>

More specifically, our contributions are:

- **LAMP Method.** We propose a data-efficient framework for parameter-controlled 3D mesh generation that aligns exemplar-specific SDF decoders and synthesizes new shapes through affine weight-space mixing.
- **Safety Metric.** We introduce a linearity-mismatch metric that identifies whether interpolated or extrapolated weight combinations remain geometrically valid.
- **Engineering Applications.** We demonstrate LAMP on DrivAerNet++ cars, BlendedNet aircraft, and industry-level vehicle families, showing controlled interpolation, large-range extrapolation, and performance-driven optimization.

2. Related Work

Generative Models for 3D Shapes. A wide range of 3D generative models have been proposed, spanning voxel grids (Wu et al., 2016), point clouds (Achlioptas et al., 2018), meshes (Groueix et al., 2018), and neural implicit representations such as signed distance functions (SDFs) (Park et al., 2019; Chibane et al., 2020; Mescheder et al., 2019; Mildenhall et al., 2021). Neural SDFs capture high-resolution geometry and have been applied to generation and reconstruction (Jiang et al., 2020) and reconstruction (Atzmon & Lipman, 2020). Recent diffusion-based 3D generators operate on implicit or latent representations, including LION, GET3D, Diffusion-SDF, SDFusion, and SALAD (Vahdat et al., 2022; Gao et al., 2022; Chou et al., 2023; Cheng et al., 2023; Koo et al., 2023). However, these methods typically assume abundant training data and lack explicit mechanisms for parameter-constrained generation or safe extrapolation. More recently, HyperDiffusion (Erkoç et al., 2023) modeled the distribution of overfit implicit networks directly in weight space, sampling new fields via diffusion. These works treat networks themselves as data points in parameter space (Shue et al., 2023; Dupont et al., 2022), but focus on unconditional sampling or learned meta-combination rather than explicit affine mixing under interpretable constraints.

Controllable and Conditional Generation. Efforts to introduce control often rely on conditioning on labels or attributes (Gao et al., 2019; Niemeyer et al., 2020), or on parametric templates derived from CAD data (Yumer & Mitra, 2016; Wu et al., 2021). Diffusion models have recently been adapted for class-conditional and partially conditional shape generation (Vahdat et al., 2022; Fei et al., 2025; Zhang et al., 2024). Multimodal conditioning has also been explored for controllable 3D generation, e.g., CLIP-Forge and Michelangelo (Sanghi et al., 2022; Zhao et al., 2023). However, these methods rarely support precise parameter specification (e.g., generating a car body with fixed ramp angle and width) and are not designed for data-efficient regimes. Our approach complements this line of work by enabling direct control through interpretable parameters.

Weight-Space Interpolation and Model Merging. A few works have shown that linear operations in weight space can produce coherent outputs (Ilharco et al., 2022). Deep Network Interpolation (DNI) (Wang et al., 2019) demonstrated smooth visual transitions by averaging parameters of two correlated image translation networks. In classification, model soups (Wortsman et al., 2022) average fine-tuned networks to improve robustness (Matena & Raffel, 2022). In generative modeling, researchers have merged GANs trained on different categories to yield hybrid semantics (Avrahami et al., 2022), and also commonly combine diffusion models by interpolating or adding weight deltas to merge styles (Biggs et al., 2024). These approaches confirm

that weight-space mixing can yield meaningful interpolations, but typically operate on pairs of models and lack mechanisms for interpretable, constraint-driven control.

Parametric Design and Engineering Constraints. Engineering design relies heavily on parametric modeling, where small sets of interpretable variables govern global shape (Seff et al., 2020; Wu et al., 2021). While recent learning-based CAD systems leverage symbolic histories or constraint graphs, they often require large, structured datasets (Rukhovich et al., 2025; Man et al., 2026). By contrast, we target the data-efficient setting where only meshes and parameter annotations are available. LAMP directly links parameters to mesh geometry via aligned SDF weight spaces, and enforces validity through a linearity-mismatch safety metric and mesh-based surrogate checks.

Shape Interpolation and Extrapolation. Latent-space interpolation has been widely explored in autoencoders (Achlioptas et al., 2018) and implicit representations (Park et al., 2019), but these spaces are often not semantically aligned, leading to unrealistic interpolations or invalid extrapolations. Our method leverages affine mixing of aligned SDF weights, which—combined with the linearity-mismatch criterion—ensures that interpolated or extrapolated meshes remain geometrically consistent and satisfy parametric constraints.

Position of This Work. Our approach bridges these threads. Like HyperDiffusion, we treat overfit exemplar networks as aligned points in parameter space, but instead of learning a generative model over them, we provide direct, interpretable control by solving for mixing coefficients in parameter space and applying them in weight space. Unlike latent traversal or disentanglement, we do not rely on a single model to encode all variation. And unlike prior weight interpolation methods, we generalize beyond two-model blends to a bank of exemplars, enabling constraint-driven, multi-way affine mixing. To our knowledge, this is the first work to formulate controllable generation as parameter-space affine mixing of aligned exemplar networks.

3. Method

We present **LAMP**, a data-efficient framework for controllable 3D mesh generation that can safely interpolate and extrapolate in parameter space. LAMP (i) constructs an *aligned* weight-space basis by overfitting signed distance function (SDF) networks to a small set of 3D shapes, (ii) solves a parameter-constrained *mixing* problem to synthesize new SDF weights and decode meshes, and (iii) evaluates reliability using a linearity-based safety metric and a surrogate that predicts parameters directly from generated meshes.

Problem Setup and SDF Weight Space. We are given

N exemplars, each with mesh $\mathcal{M}_i$ and parameter vector $\mathbf{p}_i \in \mathbb{R}^d$ (e.g., length, width, roof height, ramp angle). For every design, we overfit an SDF network f_{θ_i} , starting from a shared initialization at the mean design. This yields weights $\mathbf{w}_i = \theta_i \in \mathbb{R}^D$ that live in an approximately *aligned* weight space. Stacking rows gives

$$\mathbf{P} \in \mathbb{R}^{N \times d}, \quad \mathbf{W} \in \mathbb{R}^{N \times D}.$$

An arbitrary weight vector $\mathbf{w}$ is decoded into a mesh $\mathcal{M} = \text{Decode}(\mathbf{w})$ by evaluating the zero-level set of the SDF distribution on a dense voxel grid at the desired resolution, and extracting the isosurface using the marching cubes algorithm (Lorensen & Cline, 1998).

Parameter-Constrained Mixing. Given a target parameter vector $\mathbf{p}_d$ with a constrained index set $\mathcal{C}$, we solve the following optimization problem for mixing coefficients α :

$$\min_{\alpha} \|\mathbf{P}_{:, \mathcal{C}}^\top \alpha - \mathbf{p}_{d, \mathcal{C}}\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^\top \alpha = 1. \quad (1)$$

The synthesized weights and decoded mesh are

$$\mathbf{w}_d = \mathbf{W}^\top \alpha, \quad \mathcal{M}_d = \text{Decode}(\mathbf{w}_d). \quad (2)$$

Negative α is allowed, enabling extrapolation beyond dataset bounds.

Theoretical Justification of Mixing. Our framework builds on two key assumptions:

- (A1) *Linearity of the control-point map.* Each mesh $\mathcal{M}_p$ can be described by a set of geometric control points $C(p) \in \mathbb{R}^{n \times 3}$, such as spline knots or characteristic vertices that define the surface. We assume the map from parameters to control points is linear:

$$C\left(\sum_i \alpha_i p_i\right) = \sum_i \alpha_i C(p_i).$$

This reflects how most engineering deformations are modeled: affine transformations (translation, scaling, stretching), spline coefficient adjustments, or superpositions of independent deformations. Even when nonlinear parameterizations are used (e.g., quadratic variations in thickness or curvature), they can often be re-expressed in a linear basis of control-point coefficients (see Appendix A).

- (A2) *Local linearity in SDF weights.* For fixed input z , the decoder $f(z; w)$ is approximately linear in w in a neighborhood of a reference w_0 :

$$f\left(z; \sum_i \alpha_i w_i\right) \approx \sum_i \alpha_i f(z; w_i),$$

with error $O(\max_i \|w_i - w_0\|^2)$. (Proof in Appendix C)

Under (A1)–(A2), interpolating weights $\hat{w}_\alpha = \mathbf{W}^\top \alpha$ produces an SDF close to $\text{SDF}(C(\hat{p}_\alpha))$ with $\hat{p}_\alpha = \mathbf{P}^\top \alpha$, ensuring that mixing in weight space corresponds to faithful geometric interpolation and extrapolation. (Proof in Appendix B)

Safety Metric: Linearity Mismatch. We additionally quantify whether affine mixing remains in a valid linear regime (A2). For N_z sampled 3D coordinates $\{z\}$, we compute

$$\frac{1}{N_z} \sum_z \left| f(z; \sum_i \alpha_i w_i) - \sum_i \alpha_i f(z; w_i) \right|.$$

A mesh is accepted if this mismatch is below ϵ . This provides a quantitative safety threshold: low mismatch implies faithful linear mixing, while high mismatch indicates collapse (Fig. 5).

4. Experiments, Results, and Discussion

We evaluate LAMP against two representative baselines for parameter-controlled 3D generation:

DNI (Deep Network Interpolation): a learned model mapping design parameters directly to SDF decoder weights (Wang et al., 2019). **AE-LPA (Autoencoder with Latent-Parameter Alignment):** an autoencoder trained to reconstruct SDF weights with its latent subspace linearly aligned to design parameters (Jain et al., 2021).

We benchmark these methods on two recent parametric datasets (Appendix M): **DrivAerNet++**: a large-scale multimodal car dataset with $\sim 8,000$ distinct geometries, each annotated with 26 interpretable design parameters, high-resolution meshes, and CFD-based aerodynamic coefficients. (Elrefaie et al., 2024) **BlendedNet**: a blended wing-body (BWB) aircraft dataset with 999 geometries, each simulated under 9 flight conditions, and annotated with planform parameters such as chord-length ratios, spanwise widths, and sweep angles. (Sung et al., 2025)

Evaluation Metrics. For in-dataset generation (when the target mesh is available), we evaluate: **Chamfer Distance (CD)** and **Intersection-over-Union (IoU)** between generated and reference meshes (see Appendix G). **Parameter Error**: mean absolute error (MAE) between target parameters and surrogate-predicted parameters for the generated mesh. For out-of-distribution extrapolation (no ground-truth mesh), we evaluate: **Parameter Fidelity**: surrogate-predicted MAE and R^2 between target parameters p_d and inferred $\hat{p}$, **Distributional Similarity**: Minimum Matching Distance (MMD) between generated shapes and a reference set of geometries. (see Appendix G).

Constraint Compliance via Mesh-Based Surrogates. To assess whether generated shapes respect design-parameter constraints, we employ a mesh-based surrogate model

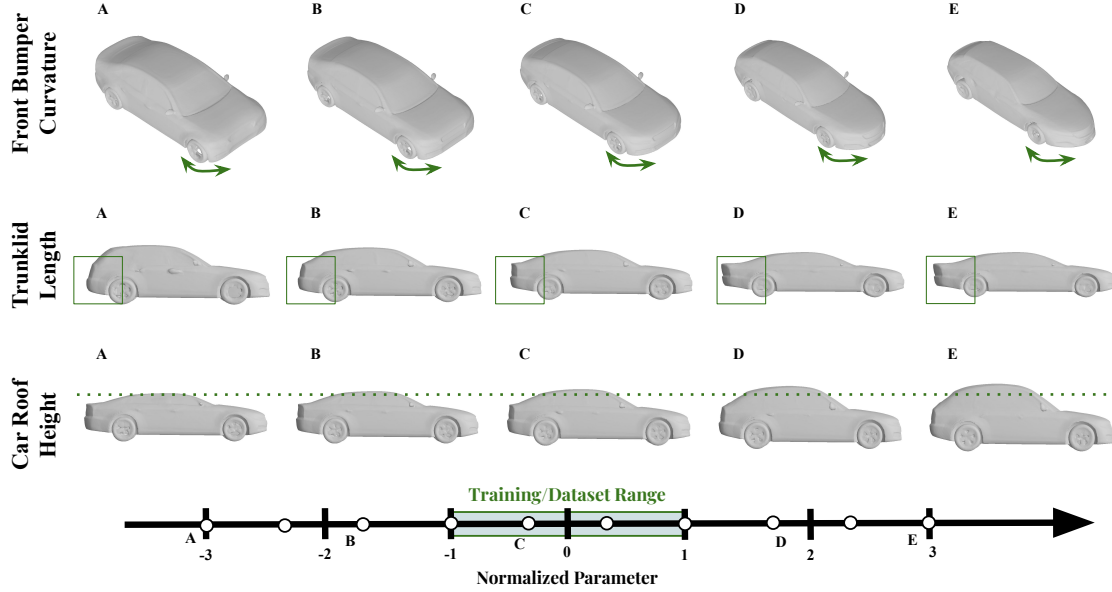


Figure 2. Single-parameter extrapolation showing LAMP’s smooth, plausible geometries.

trained to predict interpretable parameters (e.g., geometric or performance attributes). We compute fixed, randomly initialized PointNet embeddings for each decoded mesh (Amid et al., 2022), and fit a LASSO regressor (Tibshirani, 1996) to map embeddings to physical parameters. Despite using an untrained encoder, the surrogate consistently achieves $R^2 > 0.9$ on held-out test sets, providing a robust parameter validator. When possible, we further cross-check compliance through *direct geometric measurements* of the meshes (details in Appendix H).

Table 1. Interpolation results on DrivAerNet++.

Method	# Samples	CD ↓	IoU ↑ (%)	MAE ↓
DNI	100	0.0118	97.12	0.0015
AE-LPA	100	0.0181	88.21	0.0025
AE-LPA	1000	0.0144	92.63	0.0020
LAMP (Ours)	100	0.0117	97.24	0.0014

Interpolation within Dataset Range. We evaluate interpolation by reconstructing meshes from randomly sampled dataset examples that fall within the parameter range but are excluded from training. As shown in Table 1, LAMP achieves the best performance across Chamfer Distance, IoU, and parameter error. Notably, with only 100 samples, it surpasses AE-LPA trained on 1000 samples, demonstrating strong sample efficiency in low-data regimes.

Extrapolation within Dataset Range. We next evaluate extrapolation slightly beyond the training set. Training samples are drawn from a centered 50% interval of the parameter range, and evaluation is performed on cars with parameter values outside that interval. Figure 9 illustrates the *single-parameter* case, showing extrapolation along front bumper

curvature. While DNI—the strongest baseline within the dataset range—begins to drift outside the training span, LAMP maintains smooth, parameter-consistent geometry.

Table 2 reports quantitative results. For single-parameter extrapolation, LAMP improves Chamfer Distance, IoU, and surrogate error relative to DNI. The advantage grows in the *multi-parameter* setting, where three random parameters are simultaneously extrapolated outside a 60% centered range: LAMP reduces surrogate MAE by 20–30%, showing greater robustness under multiple constraints.

Table 2. Extrapolation within dataset range (DrivAerNet++). LAMP outperforms DNI in both single- and multi-parameter settings, with especially large gains when multiple parameters are extrapolated simultaneously.

Method	Single Parameter			Multi-Parameter (3)		
	CD ↓	IoU ↑	MAE ↓	CD ↓	IoU ↑	MAE ↓
DNI	0.0129	95.32	0.098	0.0139	94.28	0.186
LAMP	0.0126	95.75	0.077	0.0130	95.29	0.144

Large-Range Extrapolation Beyond the Dataset Bounds.

We next evaluate extrapolation far outside the training span, extending parameters up to $\pm 100\%$ beyond the dataset limits (i.e., three times the original parameter range). This task is challenging, as models must generate plausible geometries without support from nearby training examples. We consider two extrapolation settings:

- 1. Single-Parameter Extrapolation.** For each parameter, we sweep across the extrapolated range ($3\times$ the dataset span) using 10 uniformly sampled target values, while

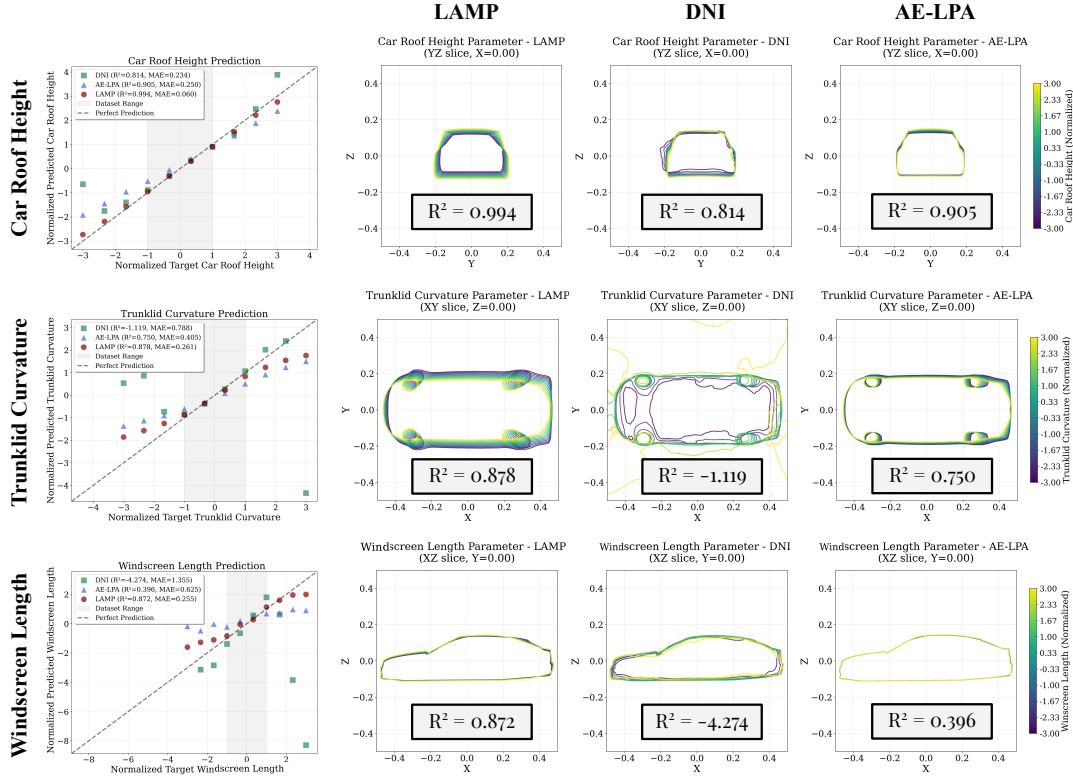


Figure 3. Single-parameter extrapolation beyond the dataset range, with all other parameters allowed to vary. Left: surrogate-predicted vs. target parameters. Right: decoded cross-sections. LAMP extrapolates smoothly, while DNI collapses and AE-LPA fails to reach the expected parameter range.

allowing all other parameters to vary freely. This setup evaluates whether models sustain smooth and consistent shape evolution along a single direction of variation (Figs. 3, 2).

2. **Multi-Parameter Extrapolation (4D).** We repeat 100 trials where four parameters are randomly selected and set outside the dataset range (up to 50% extrapolation, i.e., $2\times$ the span). Each method generates meshes under these conditions, which are then converted to point clouds, embedded with a fixed PointNet encoder, and visualized in 2D via multidimensional scaling (MDS). This reveals both fidelity and diversity of extrapolated generations (Fig. 4).

Results. Quantitative results are reported in Table 3. For single-parameter extrapolation, LAMP reduces parameter error by more than 40% compared to DNI and achieves $R^2 = 0.90$ versus $R^2 = 0.14$ for DNI. In the four-parameter case, DNI collapses completely ($R^2 < 0$), AE-LPA remains confined to the convex hull of the dataset, while LAMP sustains high fidelity ($R^2 = 0.87$) with low MMD and surrogate error. Figures 2 and 3 illustrate the single-parameter sweeps: DNI collapses outside the training range and AE-LPA undershoots, whereas LAMP produces smooth, parameter-consistent variations that remain geometrically

valid across the entire sweep. In the more challenging four-parameter extrapolation (Fig. 4), DNI collapses to invalid meshes and scatters randomly in embedding space, AE-LPA stays trapped in the dataset’s convex hull with low diversity, while LAMP extrapolates beyond the convex hull and generates high-fidelity meshes in previously unobserved regions. This shows that LAMP can be used for dataset augmentation and controlled generation from a few samples.

Table 3. Large-range extrapolation on DrivAerNet++. LAMP sustains high fidelity, while DNI collapses in multi-parameter extrapolation.

Method	Single Parameter			Multi-Parameter (4)		
	MMD ↓	MAE ↓	R^2 ↑	MMD ↓	MAE ↓	R^2 ↑
DNI	0.043	0.705	0.143	0.060	1.313	-5.768
AE-LPA	0.031	0.405	0.750	0.030	0.420	0.685
LAMP	0.030	0.247	0.902	0.030	0.324	0.867

Challenges and Safety in Extrapolation with Limited Data. A major challenge of large-range extrapolation is validating the plausibility of generated geometries when training data are scarce. With only 100 samples, mesh-based surrogates cannot be trained reliably to evaluate out-of-distribution designs. In such low-data regimes, models

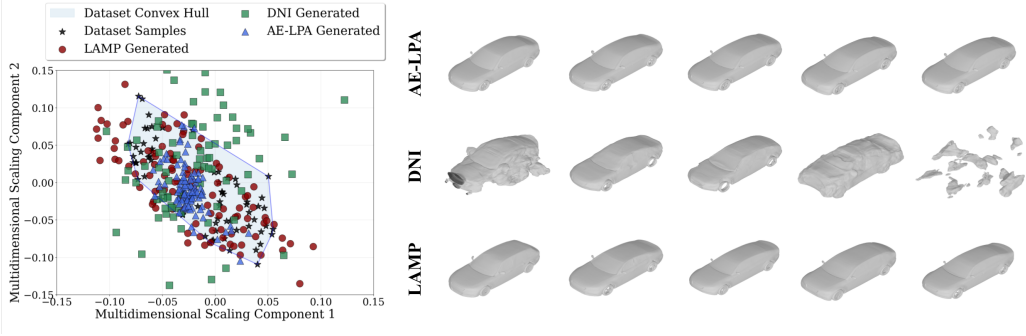


Figure 4. Four-parameter extrapolation. Left: distribution of generated meshes in a 2D point cloud embedding. Right: decoded examples. LAMP remains within plausible regions, DNI collapses to invalid meshes, and AE-LPA remains stuck in the dataset convex hull, lacking diversity.

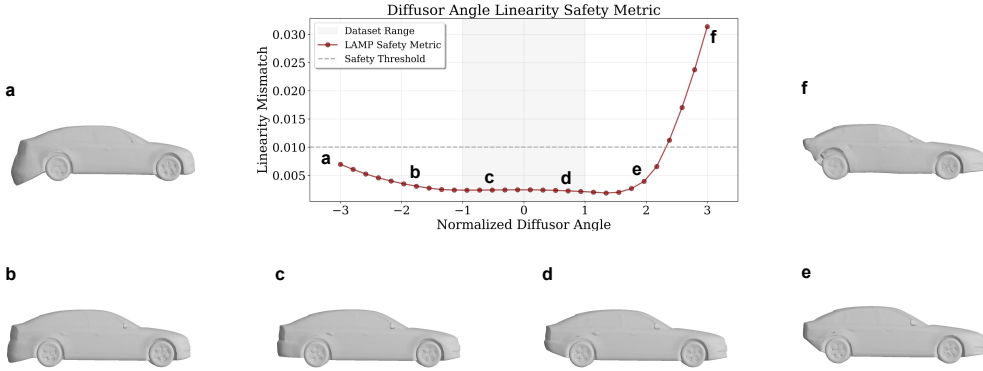


Figure 5. Linearity-mismatch safety metric for diffuser angle extrapolation. Failures (e.g., sample f) occur when the metric exceeds the threshold. See Appendix K for more.

may occasionally produce collapsed or implausible shapes, especially when extrapolating far beyond the dataset span.

To address this, we introduce a *linearity-mismatch safety metric* (Sec. 3), which quantifies whether affine weight mixing remains locally valid in SDF space. Unlike surrogate-based validation, this metric is lightweight and data-independent, enabling it to flag unsafe generations even when labeled training data are unavailable. As shown in Fig. 5, failure cases (f) arise precisely when the mismatch score exceeds a threshold. We validate this metric against a human-annotated dataset of valid and invalid meshes (Appendix K). The results show excellent discriminative power (ROC AUC = 0.989, PR AUC = 0.990), and $\epsilon = 0.01$ emerges as a reliable threshold for separating valid from invalid generations. Additional examples of failure cases and validation analysis are provided in Figs. 19–20.

Another limitation is that we were only able to hard-code the measurement of car length, as explained in Appendix H, achieving $R^2 = 0.999$ under $\pm 100\%$ extrapolation sweeps. In contrast, other parameters were far more difficult to hard-code due to the complexity of their deformations across

different car geometries and the absence of reliable methods for accurate measurement. We include an extended limitation section in Appendix J.

Finally, we study how reliability scales with data availability (Appendix E). As shown in Table 9, increasing the training set size improves both predictive accuracy (R^2 , MAE) and the mean safe extrapolation range, which grows from $\sim 145\%$ with 10 samples to over 400% with 1000 samples before saturating. This ablation highlights both the limitations of extrapolation in extremely low-data regimes and the safety metric’s role in flagging unreliable extrapolations.

Transferability Across Aircraft and Industry-Level Vehicle Families. We test whether LAMP transfers beyond one car family by adding BlendedNet aircraft and three real-world vehicle classes: sports cars (Toyota Supra), SUVs (Toyota C-HR), and convertibles (Porsche Carrera GT). BlendedNet probes cross-domain transfer to blended wing-body aircraft; the car study probes cross-family transfer within vehicles. For the car study, we avoid learned surrogates and measure ramp angle, diffuser angle, length, width, and greenhouse angle directly on decoded meshes, testing

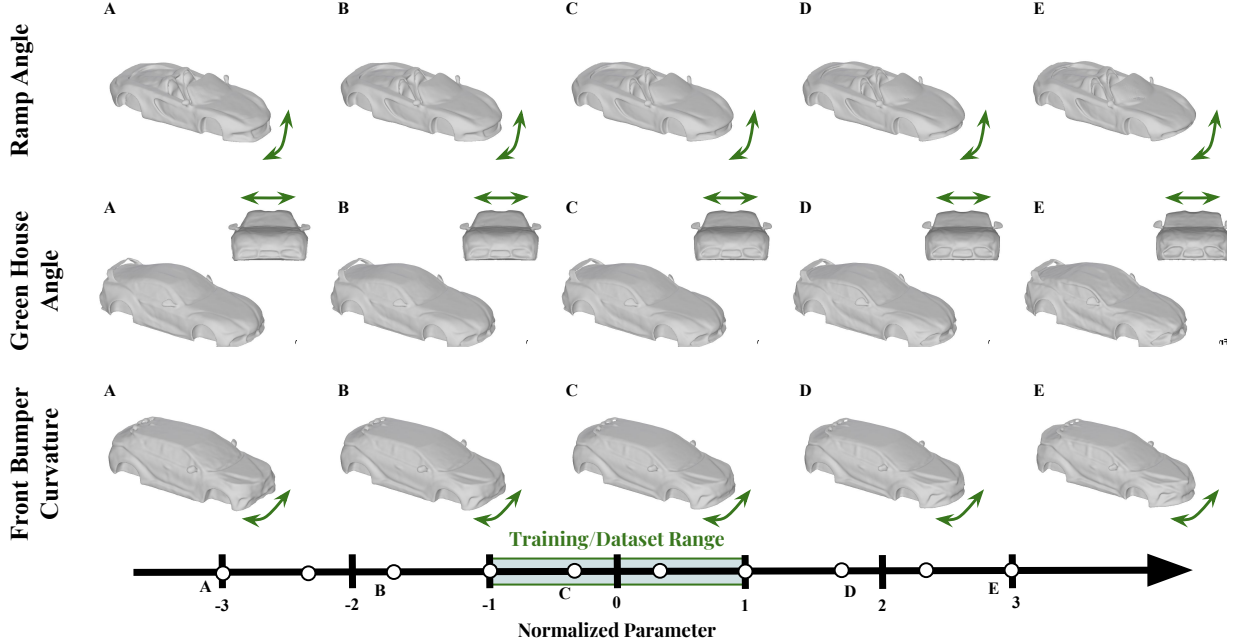


Figure 6. Industry-level car extrapolation across ramp angle, greenhouse angle, and front bumper curvature. Green marks the training range; LAMP yields smooth, plausible edits far beyond it.

true geometric control.

Table 4. Cross-vehicle extrapolation on three real-world vehicle classes with 50 samples each. Parameter accuracy uses direct mesh measurements. MMD is reported as $\times 10^3$.

Vehicle	Metric	3DS2VS (cond.)	DNI	AE-LPA	LAMP (Ours)
Sports car	MAE ↓	2.76	1.70	1.55	0.25
	R^2 ↑	-3.23	-0.06	0.20	0.97
	MMD ↓	5.50	0.50	0.49	0.48
SUV	MAE ↓	3.20	2.89	1.21	0.16
	R^2 ↑	-3.57	-0.24	0.64	0.99
	MMD ↓	3.30	0.74	0.73	0.72
Convertible	MAE ↓	2.80	3.42	1.28	0.36
	R^2 ↑	-2.12	-0.37	0.50	0.93
	MMD ↓	12.00	0.78	0.77	0.76

Tables 7 and 8 show that LAMP transfers to BlendedNet aircraft (Appendix D): with 100 samples, it halves CD relative to DNI (0.0172 vs. 0.0346), improves IoU to 95.35%, achieves the lowest parameter MAE, and remains accurate under extrapolation ($R^2 = 0.868$ for single-parameter and 0.782 for four-parameter extrapolation). Table 4 shows the same trend on industry-level cars with only 50 samples per class: LAMP achieves $R^2 > 0.93$ across sports cars, SUVs, and convertibles, cuts MAE by roughly 4–8 $\times$ over AE-LPA, and matches or improves DNI and AE-LPA in MMD while reducing MMD by over 10 $\times$ relative to conditioned 3DShape2VecSet (3DS2VS) (Zhang et al., 2023) on sports cars and convertibles. Together, these aircraft and car results show that LAMP is not tied to one dataset, topology family, or vehicle style; the same aligned SDF weight-space mix-

ing preserves both parameter control and geometric fidelity across low-data engineering design settings.

Performance-Driven Optimization. Beyond geometric parameters, we also test whether LAMP can enable *performance-based control*, where the goal is to optimize aerodynamic properties while constraining selected physical parameters. Specifically, we sample 100 random test examples from DrivAerNet++ outside the training set. For each example, we decay the drag coefficient (C_d) by 10% and select a random subset of physical parameters to be constrained to their original values, while treating the decayed C_d as an additional desired parameter. We then solve for mixing coefficients α that jointly satisfy these constraints. To validate the results, we use the mesh-based surrogate (App. H) to predict both physical parameters and drag coefficients from the generated meshes. The surrogate predictions are compared to ground-truth values, with predicted vs. target plots provided in the Appendix. We evaluate two objectives: (i) *parameter fidelity*, i.e. how closely the generated meshes respect the selected physical parameter constraints, and (ii) *drag fidelity*, i.e. how accurately the achieved reduction matches the 10% target. Here, *decay MAE* denotes the mean absolute error between the desired 10% decay and the observed (predicted) decay, averaged across all samples.

Results. Figure 7 and Table 5 summarize the outcomes. The scatter plot confirms that LAMP aligns strongly with the target drag decay ($R^2 = 0.792$), while DNI diverges completely ($R^2 < 0$) and AE-LPA shows weaker corre-

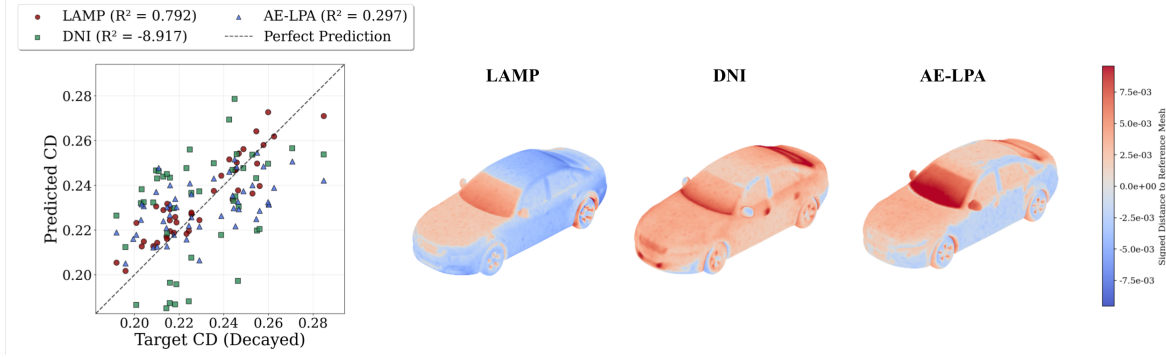


Figure 7. Performance-driven drag optimization on DrivAerNet++. Left: target vs. predicted drag reduction for LAMP, DNI, and AE-LPA. Right: error heatmaps relative to the reference mesh. LAMP achieves accurate prediction and a physically interpretable modification (flattened windscreen), while DNI and AE-LPA fail to produce aerodynamically meaningful changes.

lation. Error heatmaps highlight the geometric changes driving drag reduction: LAMP produces a visibly flatter windscreen angle, reducing flow separation and lowering drag, whereas DNI and AE-LPA introduce noisy or less interpretable deformations.

Table 5. Performance-driven optimization on DrivAerNet++ for a 10% drag reduction target. LAMP achieves the best balance between parameter fidelity and aerodynamic performance.

Method	Physical Parameters		Drag Coefficient		
	MAE ↓	R^2 ↑	Decay MAE ↓ (%)	MAE ↓	R^2 ↑
DNI	0.810	-0.184	10.2	0.333	-8.917
AE-LPA	0.161	0.797	5.2	0.146	0.297
LAMP (Ours)	0.087	0.938	2.7	0.121	0.792

Quantitatively, DNI fails to satisfy both aerodynamic and parametric constraints, with large parameter drift ($\text{MAE} = 0.810$) and unstable drag predictions ($R^2 = -8.917$). AE-LPA maintains moderate parameter fidelity ($R^2 = 0.797$) but overshoots the decay target (decay $\text{MAE} = 5.2\%$). In contrast, LAMP achieves the best trade-off: parameter fidelity improves to $\text{MAE} = 0.087$ with $R^2 = 0.938$, and drag reduction error is reduced to just 2.7%, while maintaining the strongest correlation for drag. Together, these results show that LAMP not only respects parameter constraints but also identifies physically meaningful pathways for aerodynamic optimization.

Table 6. Shared initialization is critical for aligned weight-space mixing and safe extrapolation.

Condition	R^2 ↑	MAE ↓	Safe Extrapolation Range ↑
Shared Initialization	0.838	0.507	330%
No Shared Initialization	-37.77	9.16	0%

Ablation: Effect of Shared Initialization. LAMP fits all SDF decoders from a shared initialization to keep them in a common local basin where affine weight mixing is mean-

ingful. We ablate this by training each decoder from an independent seed, keeping all else fixed; Table 6 shows that removing shared initialization collapses parameter alignment and safe extrapolation, confirming that LAMP’s gains come from aligned SDF weights overfitting.

Limitations and Future Work. While LAMP enables data-efficient and interpretable parameter-controlled 3D generation, it has several limitations. First, the method assumes that all exemplars share a common topological structure and satisfy the linear control-point model in Assumption A1; using geometries with differing topology or strongly nonlinear deformations would violate this assumption and invalidate affine mixing. Second, reliable extrapolation depends on remaining within the locally linear weight-space regime, which may break under large extrapolation or insufficient exemplar coverage. Finally, LAMP may fail when target performance objectives and physical parameters impose conflicting requirements, yielding combinations that are not jointly realizable within the exemplar set. Future work will aim to extend this framework to support multiple topological families, integrate physics-aware constraints, and develop multi-modal conditioning pipelines.

5. Conclusion

We presented LAMP, a data-efficient framework for parameter-controlled 3D mesh generation that leverages affine mixing in aligned SDF weight spaces and a linearity-based safety metric. Experiments on DrivAerNet++ and BlendedNet show that LAMP outperforms conditional autoencoders and DNI across interpolation, large-range extrapolation, and performance-guided optimization, achieving reliable control with as few as 100 exemplars. The safety score provides a principled safeguard in low-data regimes, addressing a key challenge for robust generalization. LAMP advances the goal of controllable, efficient, and verifiable 3D generation, common in engineering applications.

Impact Statement

This work contributes to the advancement of machine learning methods for scientific and engineering applications. By supporting controllable generation from small numbers of examples, it expands the applicability of machine learning in low-data engineering settings. We do not anticipate significant negative societal or ethical impacts beyond standard considerations associated with automated modeling and design tools.

Acknowledgments

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A. On the Linearity of the Control-Point Map

Our theoretical justification in §3 requires assumption (A1), i.e., that the control points defining a mesh are a linear function of design parameters. Here we explain why this assumption is natural and broadly applicable.

Affine deformations. If a mesh is transformed by translation, scaling, or uniform stretching along a coordinate axis, then each control point is exactly a linear function of the corresponding parameter. For example, increasing the wing span of an aircraft by Δs simply adds Δs to the x -coordinates of the wingtip control points.

Parameterized curves and surfaces. For many design families, parameters control polynomial or spline coefficients. Since a spline curve is itself a linear combination of control points, perturbing these coefficients changes the embedding linearly in parameter space. Even nonlinear geometric trends (e.g., quadratic camber variation) can be re-expressed in a linear basis of coefficients.

Superposition of deformations. When multiple independent deformations (length, width, rotation about an axis) are applied, the resulting control-point positions are affine functions of all parameters. Thus, any convex combination of parameter vectors yields a convex combination of control-point sets, consistent with (A1).

Coverage of practical deformations. Most engineering shape variation in engineering practice can be decomposed into linear control-point operations: extrusion height, lofting length, angle of attack, or wheelbase translation are all captured. More exotic nonlinear changes (e.g., tree-like topological branching) violate (A1) but are outside the scope of our controlled parametric families.

Takeaway. Assumption (A1) is not an artificial simplification but instead reflects how engineering models are actually parameterized: the majority of mesh variations of interest in engineering design are affine in a suitable control-point basis. This ensures that our interpolation scheme faithfully reproduces the geometry implied by parameter mixing in nearly all practical scenarios.

B. Theoretical Justification of SDF Weight Interpolation

Theorem. Let $p_1, \dots, p_N \in \mathbb{R}^d$ be parameter vectors defining meshes $\mathcal{M}_{p_i}$ via control points $C(p_i) \in \mathbb{R}^{n \times 3}$, where the map $x \mapsto C(x)$ is linear. Let $w_i \in \mathbb{R}^m$ denote the weights of an MLP SDF decoder $f(z; w)$ overfit to the mesh $\mathcal{M}_{p_i}$, and trained from a shared initialization. Suppose that:

- (1) Control point interpolation is linear:

$$C\left(\sum_{i=1}^N \alpha_i p_i\right) = \sum_{i=1}^N \alpha_i C(p_i), \quad \sum_{i=1}^N \alpha_i = 1.$$

- (2) Each MLP decoder satisfies:

$$f(z; w_i) \approx \text{SDF}(z; C(p_i)) := d_i(z),$$

where $d_i(z)$ denotes the signed distance from a queried location z to mesh $\mathcal{M}_{p_i}$.

- (3) The decoder $f(z; w)$ is locally linear in weights w for fixed input z :

$$f\left(z; \sum_{i=1}^N \alpha_i w_i\right) \approx \sum_{i=1}^N \alpha_i f(z; w_i),$$

with error bounded by $O(\max_i \|w_i - w_0\|^2)$.

Then the interpolated SDF $f(z; \hat{w}_\alpha)$ approximates the signed distance function of the mesh defined by the control points $C(\hat{p}_\alpha)$, where $\hat{p}_\alpha = \sum_i \alpha_i p_i$. That is,

$$f(z; \hat{w}_\alpha) \approx \text{SDF}(z; C(\hat{p}_\alpha)).$$

Proof. By assumption (2), for each i ,

$$f(z; w_i) \approx d_i(z) = \text{SDF}(z; C(p_i)).$$

Then by local linearity of f in weights (3),

$$f(z; \hat{w}_\alpha) \approx \sum_i \alpha_i f(z; w_i) \approx \sum_i \alpha_i d_i(z).$$

Now, because control points interpolate linearly by assumption (1), we define:

$$C_\alpha := \sum_i \alpha_i C(p_i) = C\left(\sum_i \alpha_i p_i\right) = C(\hat{p}_\alpha).$$

If the SDFs $d_i(z)$ correspond to shapes with shared topology and smooth variation in geometry, then the signed distances satisfy:

$$\sum_i \alpha_i d_i(z) \approx \text{SDF}(z; C_\alpha).$$

Therefore,

$$f(z; \hat{w}_\alpha) \approx \text{SDF}(z; C(\hat{p}_\alpha)).$$

Thus, the zero-level set of $f(z; \hat{w}_\alpha)$ corresponds to the mesh defined by $\hat{p}_\alpha$, completing the proof.

C. Approximate Linearity of the SDF Decoder in Weights

Let $z \in \mathbb{R}^3$ be a fixed 3D input point, and let $\gamma(z) \in \mathbb{R}^D$ denote its Fourier positional encoding, defined as:

$$\gamma(z) = [z, \sin(2^0 \pi z), \cos(2^0 \pi z), \dots, \sin(2^L \pi z), \cos(2^L \pi z)].$$

Let $f(z; w)$ be a feedforward multilayer perceptron (MLP) with parameters w and input $\gamma(z)$. The network is composed of K layers with weights and biases $\{W_k, b_k\}_{k=1}^K$, where:

$$\begin{aligned} h_0 &= \gamma(z), \\ h_k &= \phi(W_k h_{k-1} + b_k), \quad \text{for } k = 1, \dots, K-1, \\ f(z; w) &= W_K h_{K-1} + b_K, \end{aligned}$$

with $\phi(\cdot)$ a fixed elementwise nonlinearity (e.g., ReLU). The parameter vector w collects all $\{W_k, b_k\}$.

Claim. For fixed z , if all weights $\{w_i\}_{i=1}^N$ lie in a small neighborhood of a reference w_0 , then $f(z; w)$ is approximately linear in w . In particular, for convex coefficients $\{\alpha_i\}_{i=1}^N$ with $\sum_{i=1}^N \alpha_i = 1$, we have

$$f\left(z; \sum_{i=1}^N \alpha_i w_i\right) \approx \sum_{i=1}^N \alpha_i f(z; w_i),$$

with an error term of order $O(\|w_i - w_0\|^2)$.

Proof. Fix the input z , so $\gamma(z)$ is constant. Consider the Taylor expansion of $f(z; w)$ about w_0 :

$$f(z; w) = f(z; w_0) + \nabla_w f(z; w_0)^\top (w - w_0) + R(w),$$

where $R(w)$ is the second-order remainder term.

Applying this to each w_i gives

$$f(z; w_i) = f(z; w_0) + \nabla_w f(z; w_0)^\top (w_i - w_0) + R(w_i).$$

Now evaluate at the convex combination $\hat{w}_\alpha = \sum_i \alpha_i w_i$:

$$f(z; \hat{w}_\alpha) = f(z; w_0) + \nabla_w f(z; w_0)^\top \left(\sum_i \alpha_i (w_i - w_0)\right) + R(\hat{w}_\alpha).$$

On the other hand, the convex combination of outputs is

$$\sum_i \alpha_i f(z; w_i) = f(z; w_0) + \nabla_w f(z; w_0)^\top \left(\sum_i \alpha_i (w_i - w_0) \right) + \sum_i \alpha_i R(w_i).$$

Subtracting the two expressions gives

$$f(z; \hat{w}_\alpha) - \sum_i \alpha_i f(z; w_i) = R(\hat{w}_\alpha) - \sum_i \alpha_i R(w_i).$$

Since the remainder terms $R(\cdot)$ are second-order in the deviations $(w_i - w_0)$, this difference is $O(\max_i \|w_i - w_0\|^2)$. Thus, when all weights are close, the error is small and the decoder behaves approximately linearly in w .

D. Additional Quantitative and Qualitative Results

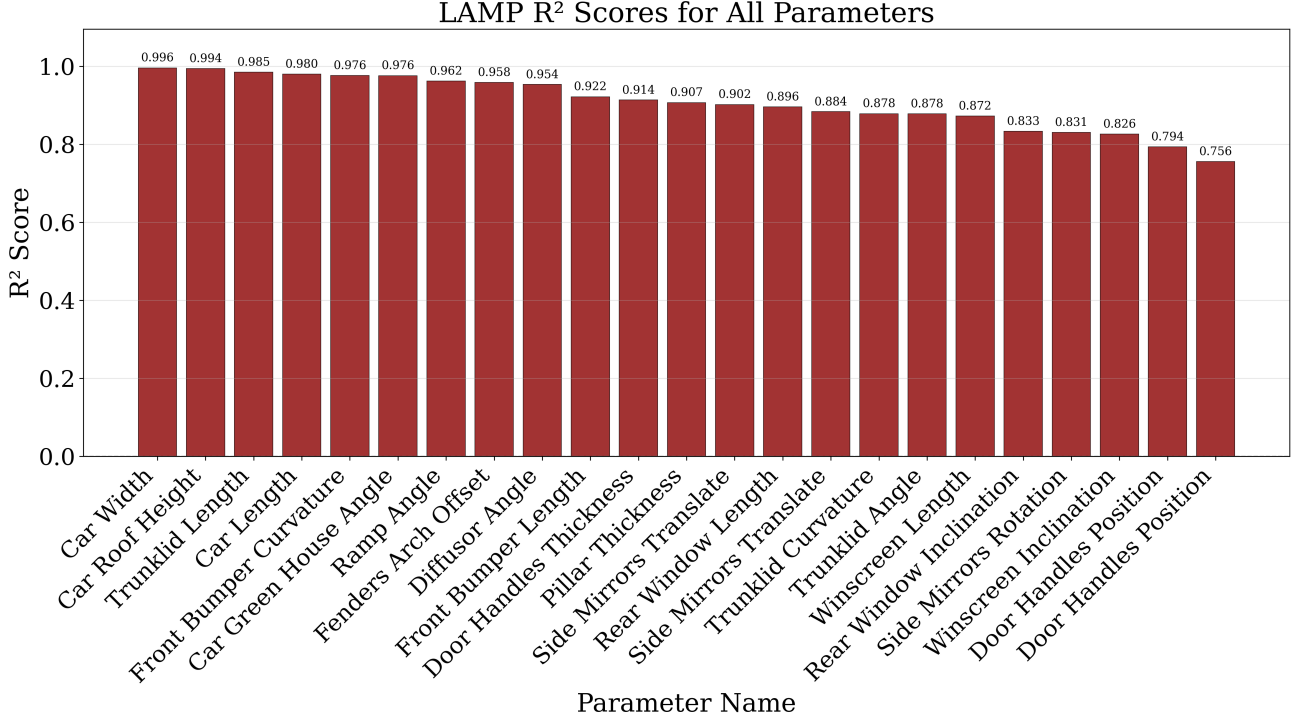


Figure 8. LAMP’s R^2 scores for single-parameter sweeps on the DrivAerNet++ dataset, extrapolated $\pm 100\%$ beyond the dataset range.

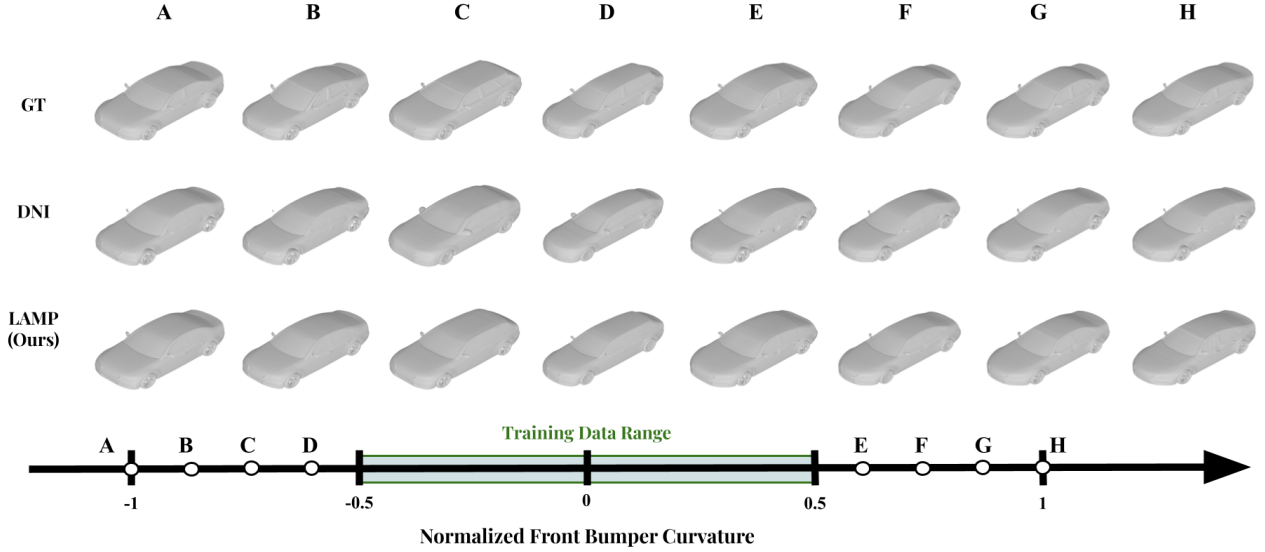


Figure 9. Single-parameter extrapolation within dataset range on DrivAerNet++. Training samples are restricted to a centered 50% interval of the parameter, while evaluation is performed outside this interval. LAMP maintains smooth, plausible extrapolation, while DNI drifts away from target shapes.

Table 7. Interpolation performance. We compare LAMP against DNI and AE-LPA baselines on BlendedNet using Chamfer Distance (CD ↓), Intersection-over-Union (IoU ↑, in %), and surrogate-based parameter error (↓). Training uses 100 samples from the dataset, and testing uses 200 held-out samples.

Dataset	Method	# Samples	CD ↓	IoU ↑ (%)	MAE ↓
BlendedNet	DNI	100	0.0346	94.21	0.0038
	AE-LPA	100	0.0393	88.26	0.0078
	LAMP (Ours)	100	0.0172	95.35	0.0031

Table 8. Large-range extrapolation (BlendedNet) up to $\pm 50\%$. LAMP sustains high fidelity ($R^2 > 0.78$) for both single- and multi-parameter extrapolation. Training uses 100 samples from the dataset. For single-parameter extrapolation, we sample 10 values uniformly per parameter within the extrapolated range, constraining that parameter while allowing the others to vary. For multi-parameter extrapolation, we repeat 100 trials where four parameters are randomly selected and set outside the dataset range (up to 50% extrapolation, i.e., twice the original span).

Dataset	Method	Single Parameter			Multi-Parameter		
		MMD ↓	MAE ↓	R^2 ↑	MMD ↓	MAE ↓	R^2 ↑
BlendedNet	DNI	0.038	0.392	0.784	0.040	0.435	0.521
	AE-LPA (100)	0.039	0.611	0.169	0.043	0.823	-0.069
	LAMP (Ours)	0.035	0.305	0.868	0.037	0.353	0.782

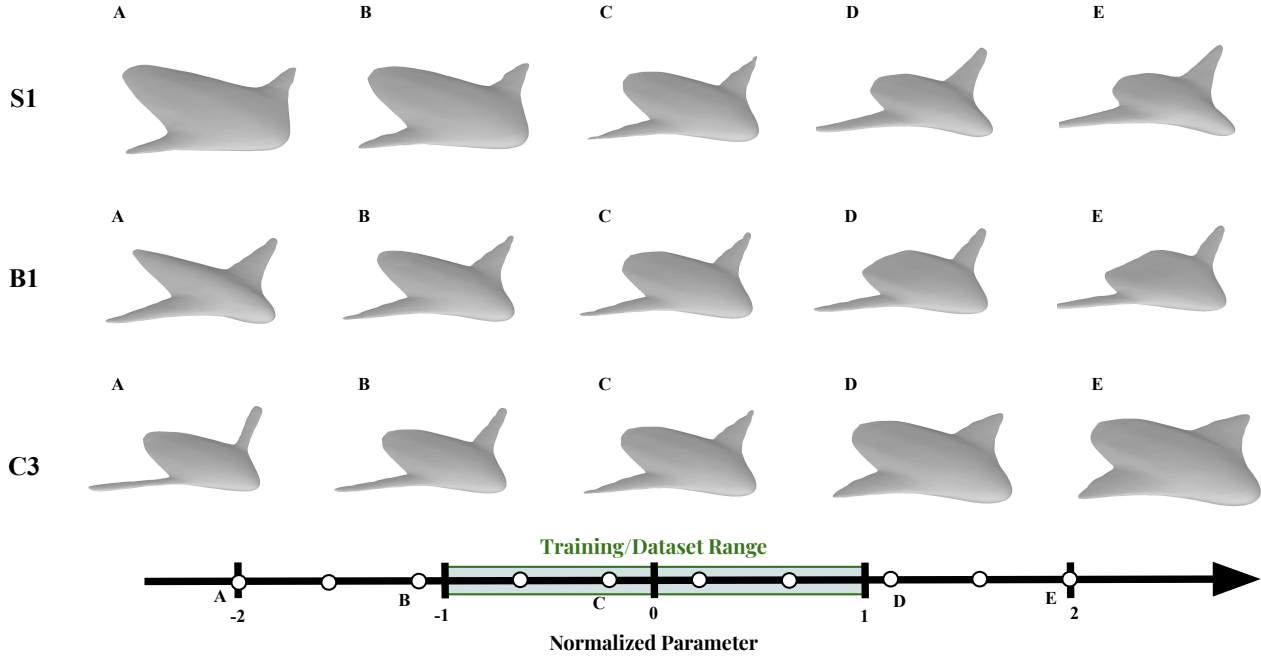


Figure 10. Single-parameter sweep on BlendedNet. LAMP sustains smooth, plausible geometry under large parameter shifts.

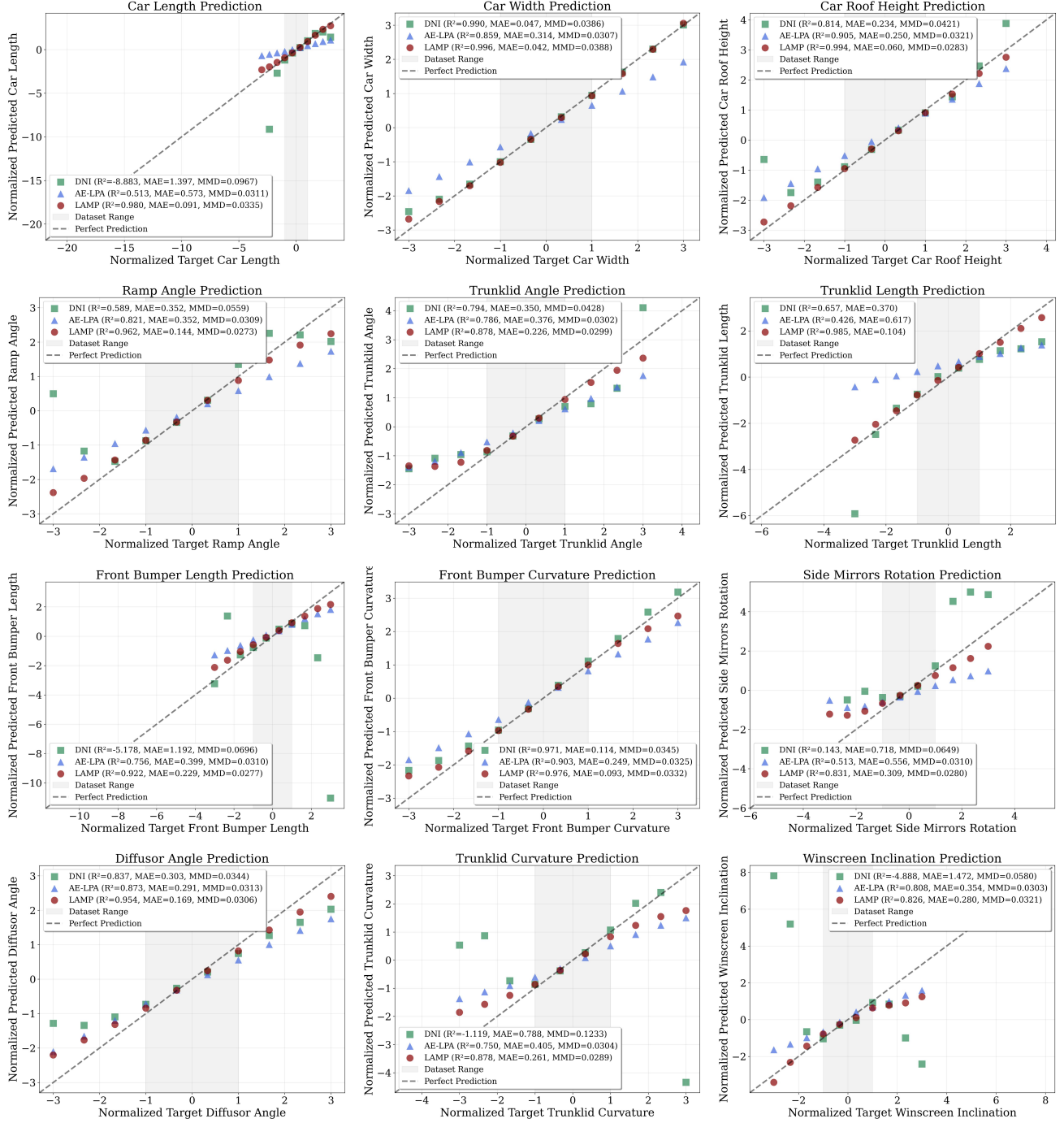


Figure 11. Single-parameter extrapolation beyond the dataset range. All other parameters are allowed to vary. The plots show surrogate-predicted versus target parameters, comparing LAMP against the baselines DNI and AE-LPA.

E. Ablation Study: How Does Sample Size Affect Reliability and Extrapolation in LAMP?

We ablate the effect of sample size in Table 9. As the number of samples increases, MAE decreases while both R^2 and the mean safe extrapolation range (%) increase, before plateauing at larger sample counts. This trend indicates that performance improves with more samples but saturates beyond a certain scale.

Table 9. Ablation study on mixing quality and reliability across different numbers of samples using LAMP

Number of Samples	$R^2 \uparrow$	MAE $\downarrow$	Mean Safe Extrapolation Range (%) $\uparrow$
10	-7.289	2.650	145.8
50	-0.214	1.083	213.9
100	0.838	0.507	330.6
500	0.849	0.479	418.1
1000	0.862	0.486	427.8

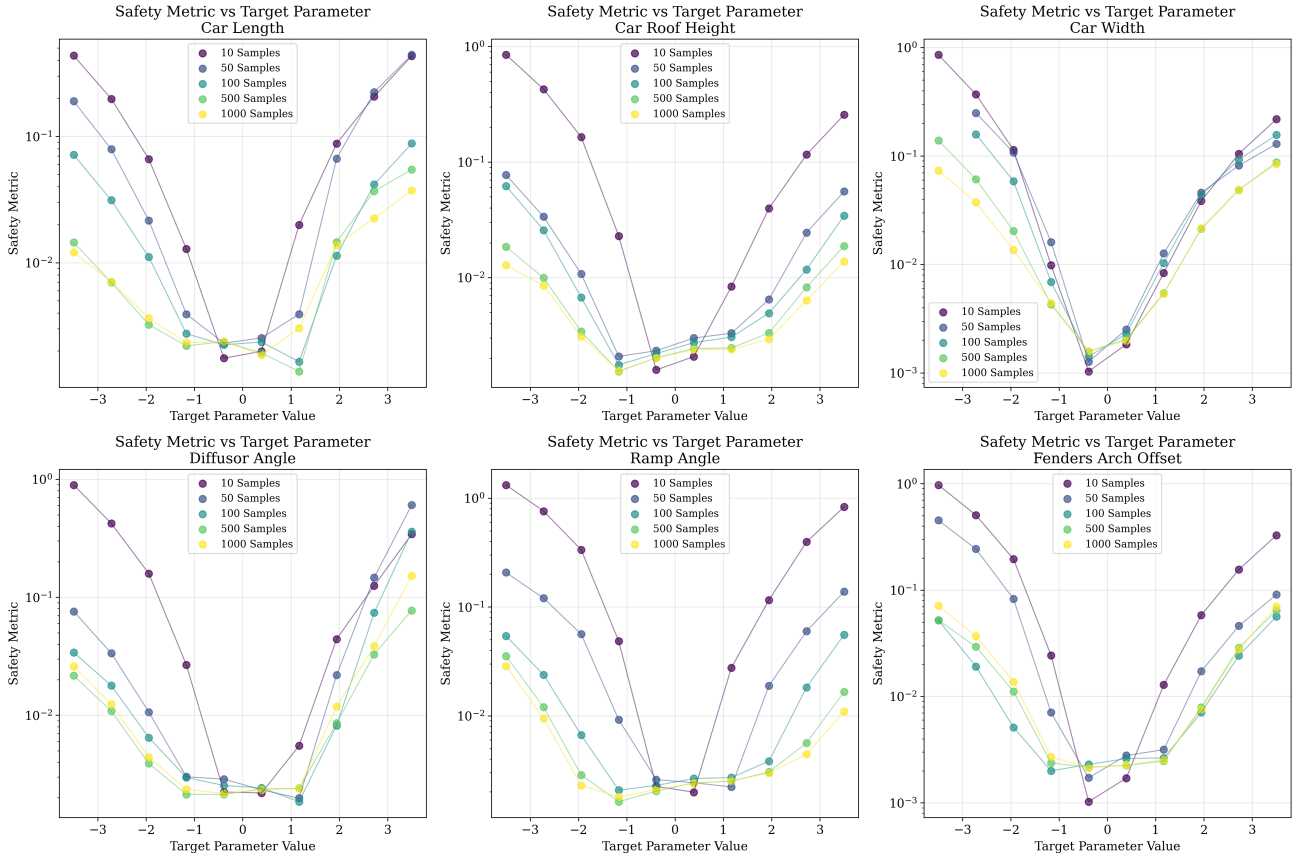


Figure 12. Safety metric values as a function of target parameter sweeps across six design parameters in DrivAerNet++. Curves correspond to different training set sizes (10–1000 samples). Larger datasets consistently reduce the safety metric, indicating more reliable extrapolation across parameter ranges.

F. Ablation Study: How Does SDF Decoder Finetuning Affect Reliability and Extrapolation in LAMP?

Sensitivity to Initialization

To evaluate whether LAMP depends on a shared initialization during SDF overfitting, we trained exemplar SDF decoders both with and without a common initialization. With a shared initialization, all exemplars remain in a common local basin, preserving strong weight-space alignment and enabling smooth, parameter-consistent interpolations.

Removing the shared initialization forces each decoder to converge to a different basin, breaking weight-space alignment. Under this setting, affine interpolation produces severely distorted and non-car-like shapes, confirming that a shared initialization is essential for maintaining a coherent basis.

Robustness to Random Seeds

We next tested whether varying the random seed (while keeping the shared initialization fixed) affects weight-space alignment. Using different seeds slightly reduces performance but does not break alignment: generated shapes remain visually consistent, parameter predictions stay stable, and the linearity assumption (A2) continues to hold.

In contrast, removing the shared initialization destroys linearity entirely and leads to invalid, severely distorted generations.

To quantify reliability, we measured R^2 , MAE, and safe extrapolation range over single-parameter sweeps using 100 exemplars. Results are summarized in Table 10.

Table 10. Effect of initialization and random seeds on reliability and extrapolation behavior in LAMP, evaluated over 100 exemplar SDF decoders.

Condition	R^2	MAE	Mean Safe Extrapolation Range
Same Seed	0.838	0.507	330%
Random Seed	0.773	0.610	304%
No Initialization	-37.77	9.16	0%

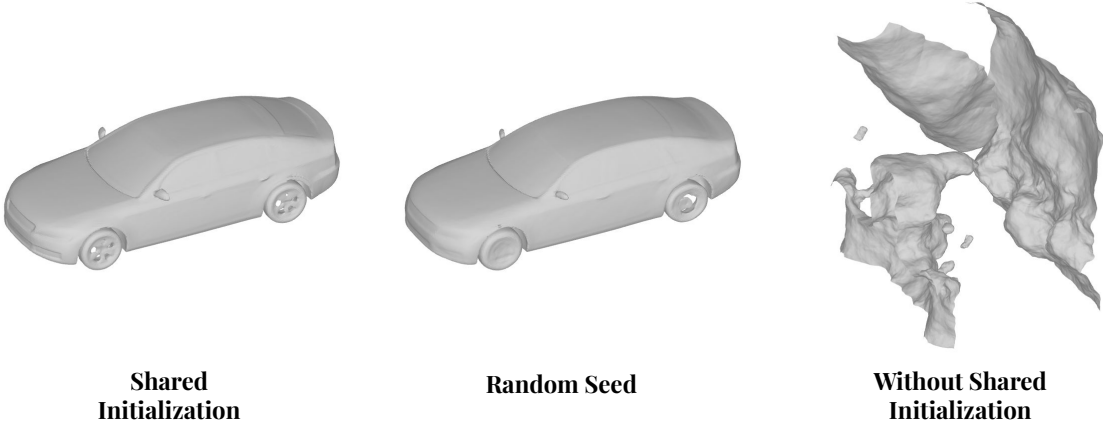


Figure 13. Representative comparison of weight-space interpolation under different training conditions: **Left:** shared initialization (stable, coherent interpolation). **Middle:** random seeds (still stable). **Right:** no shared initialization (interpolation collapses).

G. Evaluation Metrics

- **Chamfer Distance (CD):**

$$\text{CD}(X, Y) = \frac{1}{|X|} \sum_{x \in X} \min_{y \in Y} \|x - y\|_2^2 + \frac{1}{|Y|} \sum_{y \in Y} \min_{x \in X} \|y - x\|_2^2,$$

where X and Y are point clouds sampled from the predicted and reference meshes.

- **Intersection-over-Union (IoU):**

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|},$$

where A and B are voxelizations of the predicted and reference meshes.

- **Minimum Matching Distance (MMD):**

$$\text{MMD}(S_g, S_r) = \frac{1}{|S_g|} \sum_{x \in S_g} \min_{y \in S_r} d(x, y),$$

where S_g and S_r are sets of generated and reference point clouds, and $d(\cdot, \cdot)$ is typically the Chamfer distance between individual shapes.

H. Constraint Compliance Validation: Surrogates and Direct Measurements

Comparison of Mesh-Based Surrogates. In the main text, we validated design-parameter compliance using a mesh-based surrogate model: we fixed random PointNet embeddings of each decoded mesh (deterministic initialization) and fit a LASSO regressor to predict physical parameters. Interestingly, this simple surrogate achieves strong accuracy ($R^2 > 0.9$ on held-out test sets), despite the encoder being untrained.

To test whether stronger pretrained representations improve performance, we compared against the OpenShape point cloud embedding model (Liu et al., 2023). Across all parameters, the OpenShape-based surrogate achieved consistently lower R^2 scores than the randomly initialized PointNet embeddings. This suggests that domain-specific geometric structure is better captured by lightweight randomized encoders than by pretrained embeddings trained on natural 3D categories.

Figure 14 illustrates the surrogate pipeline. Figures 16 and 15 show predicted versus ground-truth parameter values on the BlendedNet and DrivAerNet++ datasets, respectively, demonstrating high accuracy across test sets. We train the surrogates on 800 samples and evaluate on a held-out test set of 200 samples.

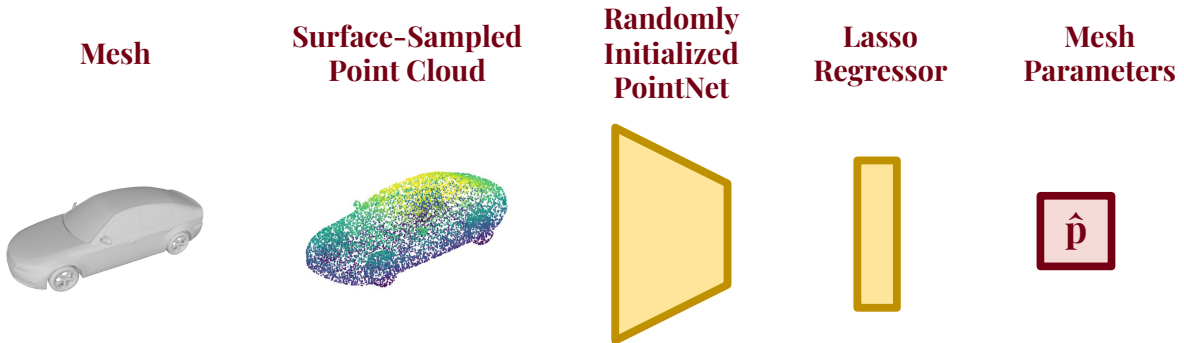


Figure 14. Diagram of the mesh-based surrogate pipeline. A decoded mesh is first converted into a surface-sampled point cloud. The point cloud is passed through a randomly initialized PointNet encoder to produce fixed embeddings, which are then mapped to interpretable mesh parameters via a LASSO regressor. This simple pipeline achieves strong predictive accuracy ($R^2 > 0.9$) despite the encoder being untrained.

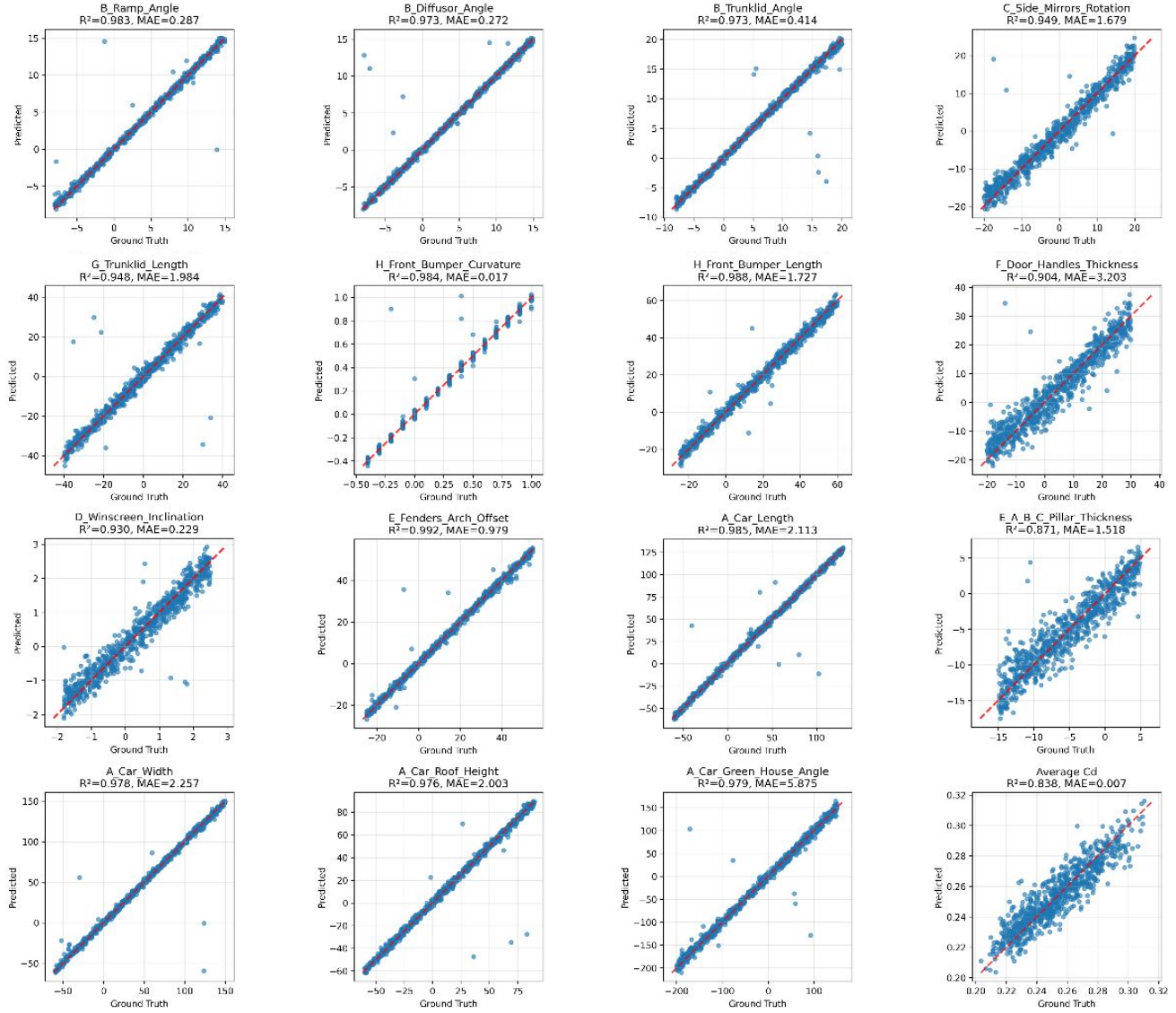


Figure 15. Predicted vs. ground truth parameters on the DrivAerNet++ test set, evaluating the mesh-based surrogate model for parameter prediction.

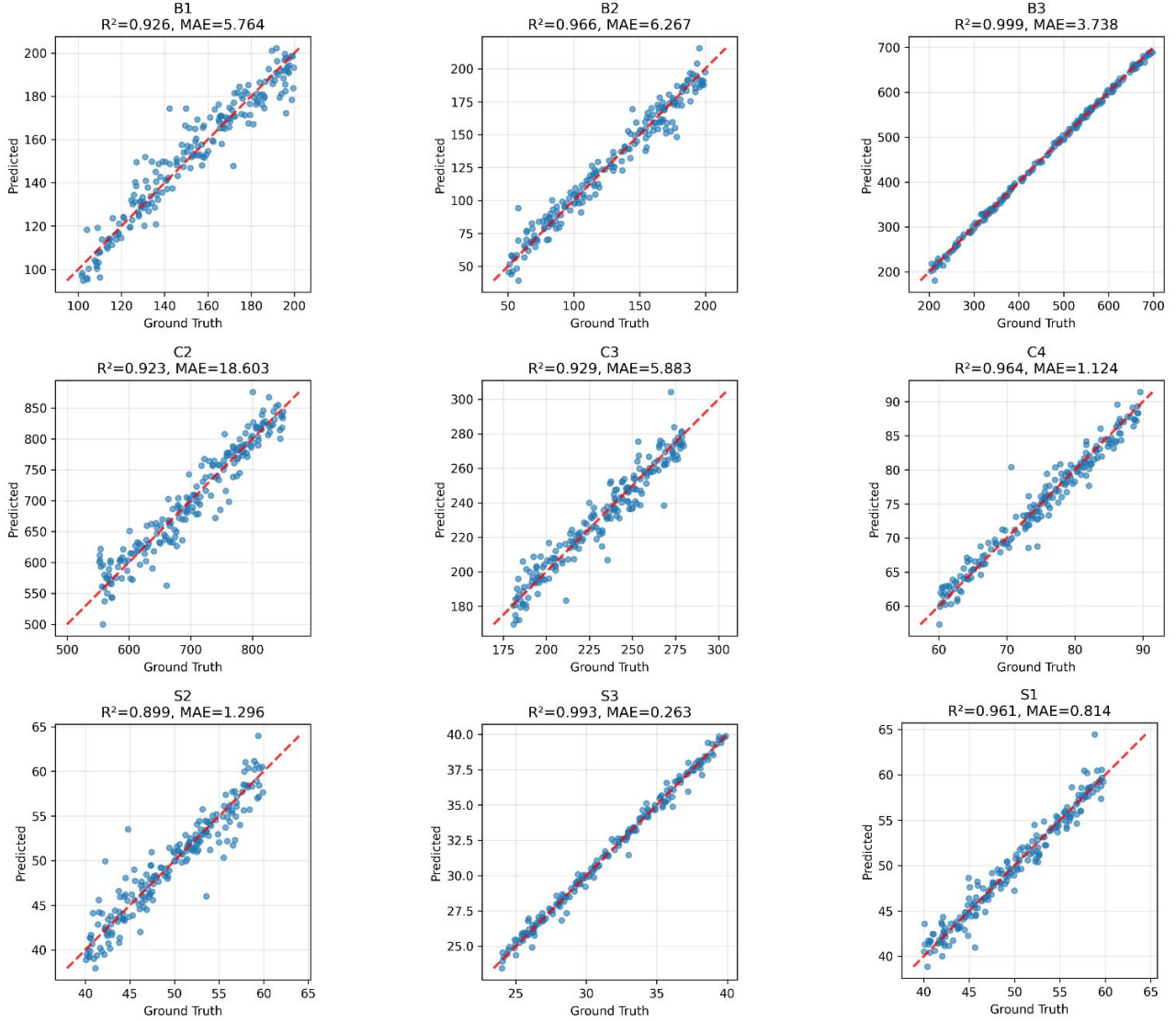


Figure 16. Predicted vs. ground truth parameters on the BlendedNet test set, evaluating the mesh-based surrogate model for parameter prediction.

Direct Geometric Measurements. Beyond surrogate-based validation, we also implemented direct geometric measurements for certain parameters. For example, to compute *Car Length* on DrivAerNet++ cars, we measure the distance between the centers of the front and rear wheels. Specifically:

1. We take slices of the decoded SDF along the wheel plane.
2. We detect circular cross-sections with radii in the expected range of wheel radii.
3. We identify the front and rear wheel centers and compute their distance.
4. We map this distance back to the labeled Car Length value by calibrating on ground-truth SDFs from the dataset.

This method provides a parameter-compliant, geometry-based validation of mesh outputs. Figure 17 shows an example of wheel detection and length estimation on generated meshes.

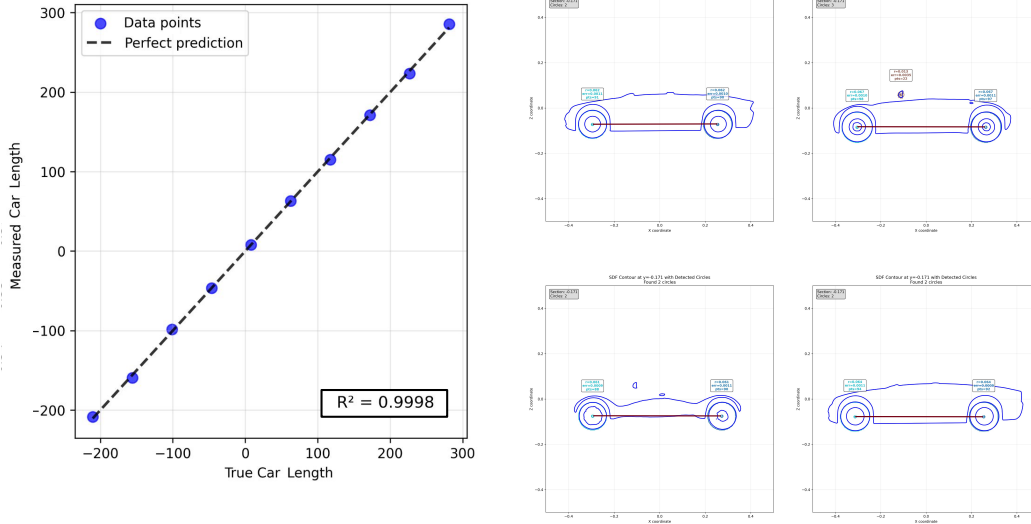


Figure 17. Direct geometric measurement of car length in DrivAerNet++. We slice the decoded SDF, detect wheel cross-sections by circle fitting, compute the distance between wheel centers, and map this measurement back to the dataset-defined Car Length parameter.

I. Runtime and Compute Cost

We report the runtime and compute cost of LAMP and the main baselines in Table 11. All methods are evaluated in the same low-data regime used in the main experiments. LAMP requires an initial per-shape SDF overfitting step, but after this preprocessing stage, generation only requires solving a small affine mixing problem and decoding the resulting SDF weights. This makes inference effectively instantaneous compared to autoencoder- or diffusion-based generators.

Table 11. Compute comparison across LAMP, baselines, and representative modern 3D generative models. LAMP has low preprocessing cost in the few-shot regime and very fast inference.

Method	Training / Preprocessing Cost	Inference Time
LAMP (Ours)	~5 min / shape; ~8.3 GPU-hours for 100 shapes	~5 ms
DNI	~8.3 GPU-hours SDF cost + ~0.1 GPU-hours training	~5 ms
AE-LPA	~10 GPU-hours	~7 s
LION (Vahdat et al., 2022)	~550 GPU-hours for ~2,500 samples	~30 s

J. Limitations of LAMP

While LAMP enables data-efficient and interpretable parameter-controlled 3D generation, it has several limitations. First, the method assumes that all exemplars share a common topological structure and satisfy the linear control-point model in Assumption A1; using geometries with differing topology or strongly nonlinear deformations would violate this assumption and invalidate affine mixing. Second, reliable extrapolation depends on remaining within the locally linear weight-space regime, which may break under large extrapolation or insufficient exemplar coverage. Finally, LAMP may fail when target performance objectives and physical parameters impose conflicting requirements, yielding combinations that are not jointly realizable within the exemplar set.

K. Validation of the Linearity-Mismatch Metric Against Human Annotated Data

Assumption. Our safety metric relies on the assumption that the decoder $f(z; w)$ is locally linear in weights w for a fixed spatial coordinate z . Formally,

$$f\left(z; \sum_{i=1}^N \alpha_i w_i\right) \approx \sum_{i=1}^N \alpha_i f(z; w_i),$$

with approximation error bounded by $O(\max_i \|w_i - w_0\|^2)$. This implies that as long as interpolations in weight space remain sufficiently close to the training exemplars, affine mixing should yield faithful mesh reconstructions. The linearity mismatch defined in Sec. 3 measures deviations from this assumption.

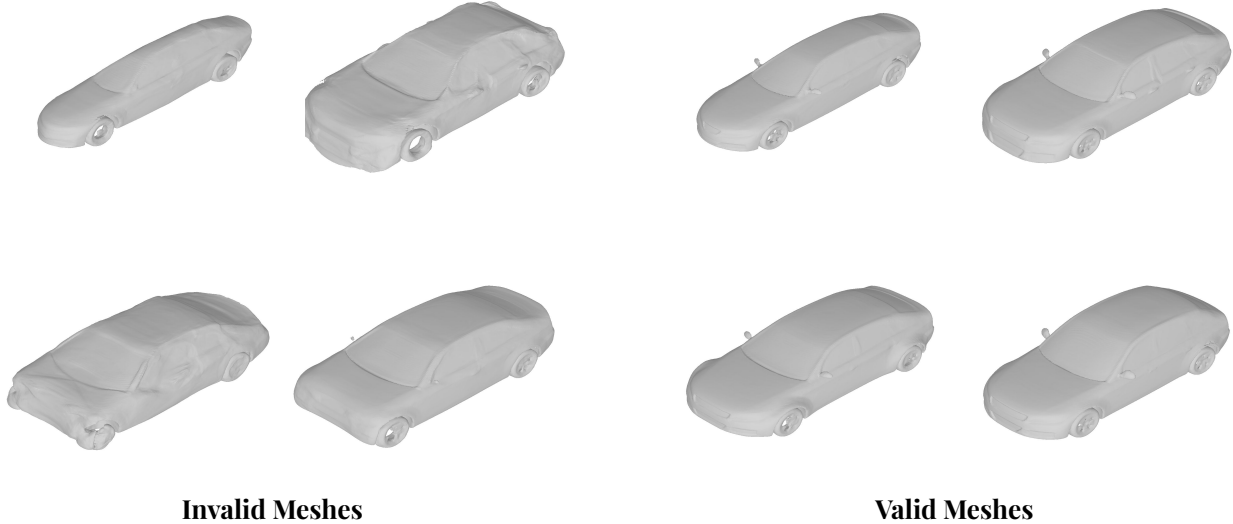


Figure 18. Examples of meshes labeled during human annotation. Left: *Invalid meshes*, which exhibit collapsed, distorted, or implausible geometries. Right: *Valid meshes*, which maintain smooth, realistic car shapes with high geometric fidelity. These labels are used as ground truth to validate the safety metric.

Dataset Construction. To empirically validate this assumption, we constructed a diagnostic dataset by systematically varying one shape parameter at a time. Each parameter was interpolated and extrapolated up to a 700% ($\pm 300\%$) increase in range compared to its span in the main dataset. For every setting, we decoded a mesh $\mathcal{M}_d$ using mixed weights $\mathbf{w}_d = \sum_i \alpha_i w_i$ and computed the linearity-mismatch score. (Fig. 18)

Human Annotation Protocol. All meshes were visually inspected and annotated as either *valid* or *invalid*. A mesh was considered valid if it was smooth and resembled a high-fidelity car geometry without collapse or severe distortion. Invalid meshes were those with degenerate or implausible deformations. This produced a binary ground-truth dataset for evaluation.

Metric Validation. We used the mismatch score to predict mesh validity and compared it against human annotations:

- The ROC curve (Fig. 20, top left) shows excellent discriminative power with an area under the curve (AUC) of **0.989**.
- The precision–recall curve (Fig. 20, top right) yields an AUC of **0.990**, indicating reliable separation of valid from invalid meshes.
- Threshold analysis (Fig. 20, bottom) reveals that $\epsilon = 0.01$ provides a good tradeoff, achieving high recall while preserving precision.

Distributional Analysis. To further assess robustness, we examined the distribution of linearity errors across mesh validity labels. As shown in Fig. 19, valid meshes cluster at low mismatch values, while invalid meshes exhibit substantially higher errors, confirming that the safety metric is well aligned with perceptual mesh quality.

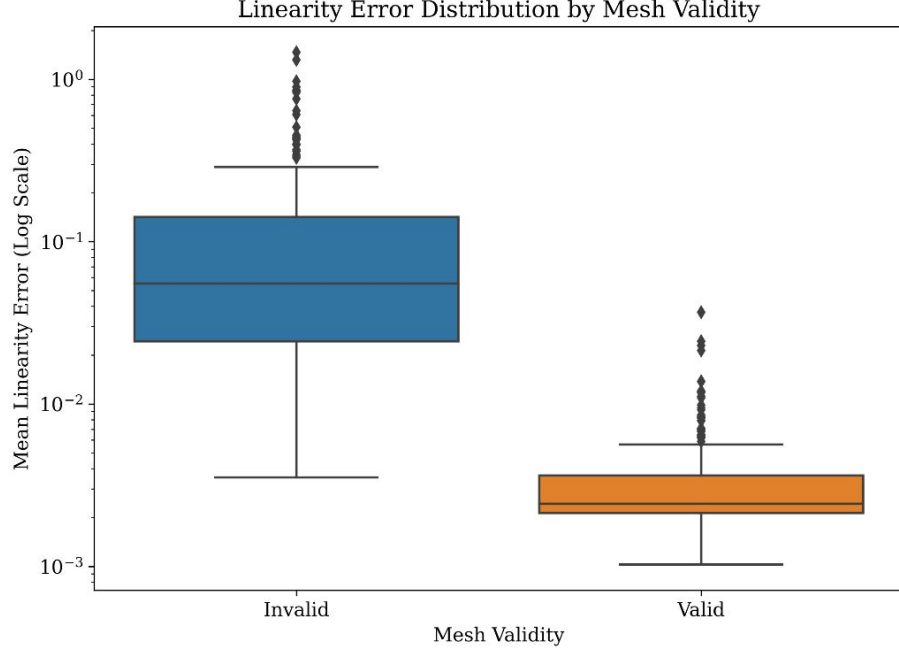


Figure 19. Box plot of mean linearity error (log scale) across meshes labeled as valid vs. invalid. Valid meshes concentrate at low mismatch values, while invalid meshes show significantly higher errors, confirming that the linearity-mismatch metric is a strong predictor of mesh validity.

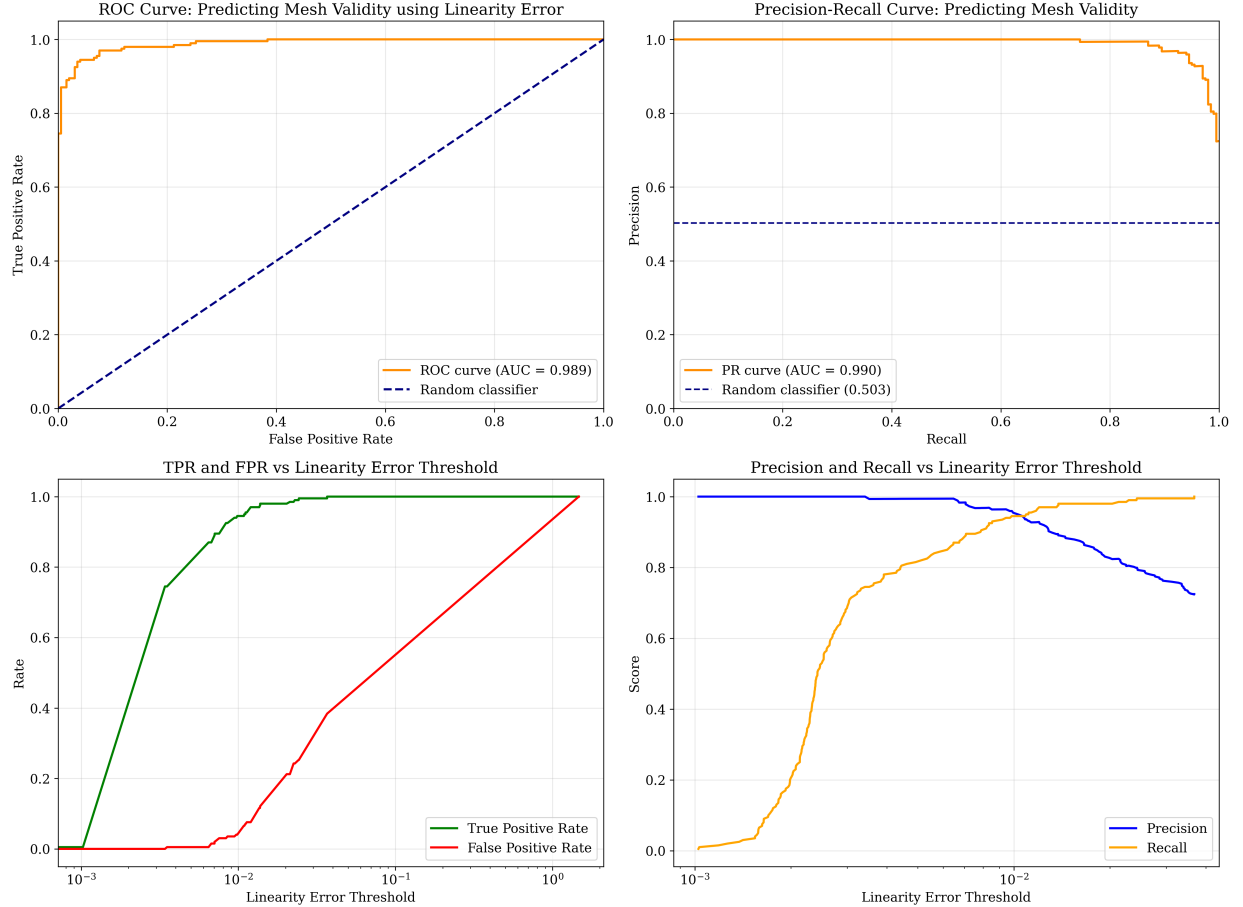


Figure 20. Validation of the linearity-mismatch safety metric against human-annotated mesh validity. Top left: ROC curve showing high discriminative power (AUC = 0.989). Top right: Precision-recall curve (AUC = 0.990). Bottom left: true positive rate (TPR) and false positive rate (FPR) as a function of linearity error threshold. Bottom right: precision and recall as a function of threshold. Together, these results confirm that the safety metric reliably predicts mesh validity, with $\epsilon = 0.01$ providing a good tradeoff between precision and recall.

Discussion. These experiments demonstrate that the linearity-mismatch metric is a reliable quantitative proxy for mesh validity. Its strong agreement with human annotations justifies our use of $\epsilon = 0.01$ as the default safety threshold throughout this work. Moreover, the threshold can be strict or relaxed depending on the application’s tolerance for geometric distortion. In aerodynamic design studies or performance-critical evaluations, we adopt a conservative cutoff to avoid any degradation in shape fidelity. In contrast, exploratory design settings may permit larger mismatch values to encourage diversity in the generated geometries. Practitioners may therefore choose a threshold smaller or larger than 0.01 depending on the demands of their downstream task.

L. Validity Ratio Under the Linearity-Mismatch Safety Filter

The main quantitative tables report results before applying the linearity-mismatch safety filter. Here, we report the fraction of generated samples that pass the safety threshold. A sample is considered valid if its linearity mismatch is below the threshold $\epsilon = 0.01$, as defined in Sec. 3. This ratio measures the practical effect of the safety filter and indicates how often LAMP’s affine weight-space combinations remain within the reliable local linear regime.

Table 12. Validity ratio of LAMP generations under the linearity-mismatch safety filter. Main-paper quantitative results are reported before filtering; this table reports the fraction of samples that pass the safety threshold.

Experiment	Valid Samples (%)
Table 1	98%
Table 2, Single Parameter	96%
Table 2, Multi-Parameter	89%
Table 3, Single Parameter	92%
Table 3, Multi-Parameter	84%
Table 4	97%

Across experiments, the majority of generated samples pass the safety filter, with validity ratios ranging from 84% to 98%. As expected, validity is highest for interpolation and single-parameter extrapolation, where affine combinations remain closer to the exemplar span. The validity ratio decreases in multi-parameter extrapolation, which imposes multiple simultaneous constraints and therefore requires larger or more complex movements in weight space.

M. Parametrization of DrivAerNet++ and BlendedNet

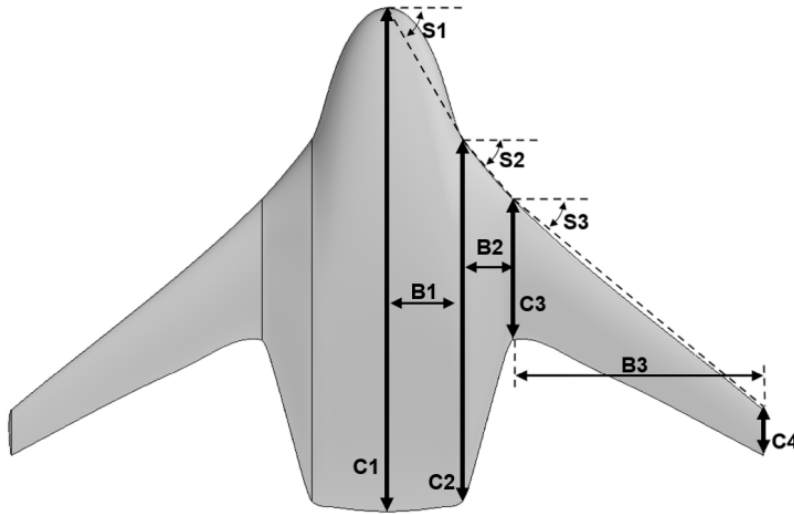


Figure 21. BlendedNet Parametrization (Sung et al., 2025)

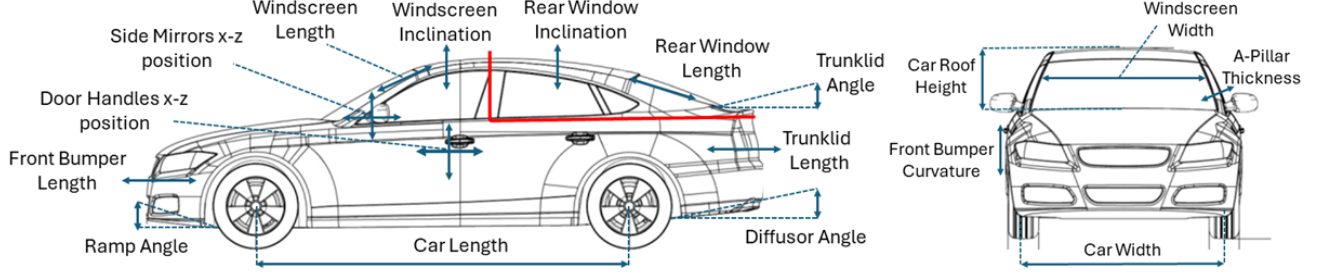


Figure 22. DrivAerNet++ Parametrization (Elrefaie et al., 2024)

N. Quantitative Validation of Linearity Assumption (A2)

To assess the linearity assumption (A2), we compared the signed distance field (SDF) predicted by the interpolated network,

$$f\left(z; \sum_i \alpha_i w_i\right),$$

against the linear combination of individual model outputs,

$$\sum_i \alpha_i f(z; w_i).$$

We sampled 10,000 random 3D query points and evaluated both quantities across a range of interpolation and mild extrapolation factors. As shown in Fig. 23, the relationship is nearly perfectly linear in-distribution, with $R^2 \approx 0.99$. Linearity degrades gradually as the extrapolation factor increases.

This empirical analysis demonstrates that SDF networks exhibit approximate linear behavior under weight-space interpolation, supporting assumption (A2).

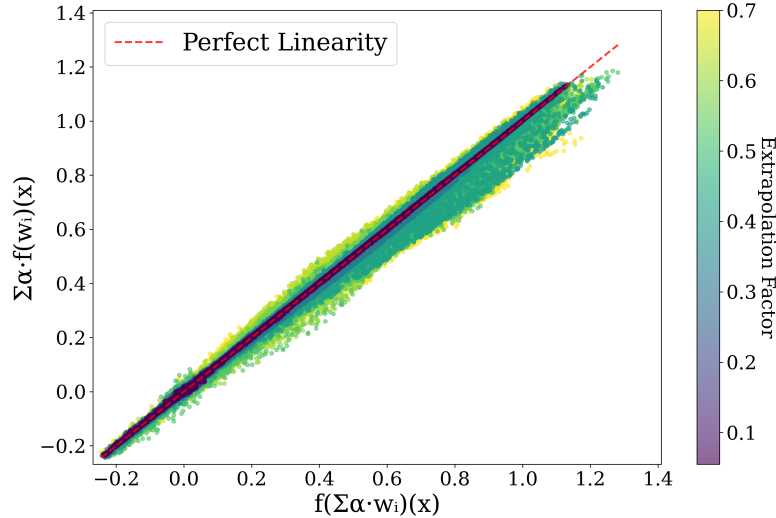


Figure 23. Linearity comparison of $f(z; \sum_i \alpha_i w_i)$ vs. $\sum_i \alpha_i f(z; w_i)$ evaluated over 10,000 3D query points while varying the extrapolation factor.

O. Extending LAMP to Non-Implicit Decoders

To examine whether LAMP can be applied beyond implicit SDF models, we performed an experiment using a non-implicit point cloud decoder that operates by deforming a spherical template into the target point cloud. First, we overfit this deformation-based decoder to a single target shape (left). We then fine-tuned the same model on a second target shape (middle), producing two distinct sets of network weights corresponding to two different geometries.

With these two trained decoders, we linearly interpolated between their weight vectors and decoded the resulting intermediate geometry. As shown in Fig. 24, the interpolated weights produce a smooth and coherent shape that lies between the two endpoints (right), confirming that weight-space interpolation remains valid even in non-implicit architectures.

This experiment demonstrates that the LAMP principle, leveraging linearity and smoothness under weight interpolation for controllable geometry manipulation, extends naturally to deformation-based point cloud decoders. This suggests that LAMP provides a general mechanism for parameterized geometry control across diverse neural shape representations.

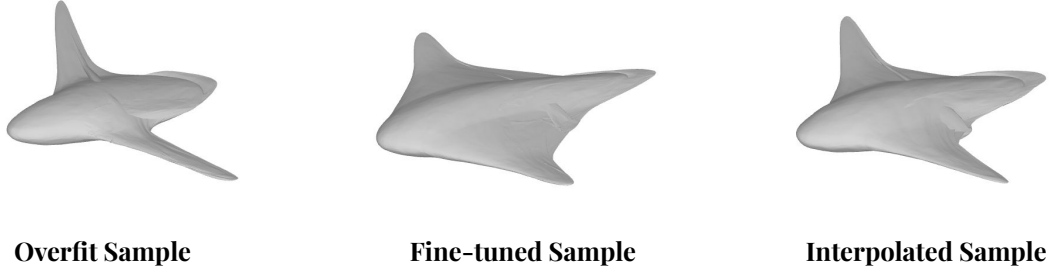


Figure 24. Interpolation using a deformation-based point cloud decoder. **Left:** model overfit to the first target (sphere deformed into shape A). **Middle:** model fine-tuned on the second target (sphere deformed into shape B). **Right:** linear interpolation in weight space yields a valid intermediate geometry, showing that LAMP extends beyond implicit SDF decoders.

P. Multi-Objective LAMP: Balancing Parameter Satisfaction and Safe Generation

Formulation

A natural extension of LAMP is to treat parameter matching and geometric reliability as a multi-objective optimization. Instead of minimizing only the parameter error, we solve

$$\min_{\alpha} \|\mathbf{P}_{:,C}^{\top} \alpha - \mathbf{p}_{d,C}\|_2^2 + \lambda \|\alpha\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^{\top} \alpha = 1. \quad (3)$$

where $\|\alpha\|_2$ serves as a fast extrapolation proxy. Evaluating the full linearity-mismatch safety metric at each iteration requires evaluating the N unflattened SDF decoders for each of the 3D query points, which is computationally heavy. In contrast, $\|\alpha\|$ varies monotonically with extrapolation and correlates strongly with the final safety score, enabling efficient exploration of accuracy vs reliability trade-offs. Since LAMP’s optimization is extremely fast, sweeping λ yields the Pareto frontier essentially for free, letting users select conservative, balanced, or exploratory solutions depending on application tolerance.

Conflicting Parameters and Safety Behavior

Conflicting or mutually incompatible parameter targets often force the optimization outside the valid linear regime, producing large $\|\alpha\|$ and correspondingly high safety scores. The leftmost example in Fig. 25 shows such a case: although the parameter error (MAE) is low, the required extrapolation produces visible geometric distortion. When decoded into meshes, these settings exhibit collapse or deformation, and the linearity-mismatch metric correctly flags them as invalid. This multi-objective formulation provides a principled way to avoid such failure modes while still permitting controlled extrapolation for creative or exploratory design tasks.

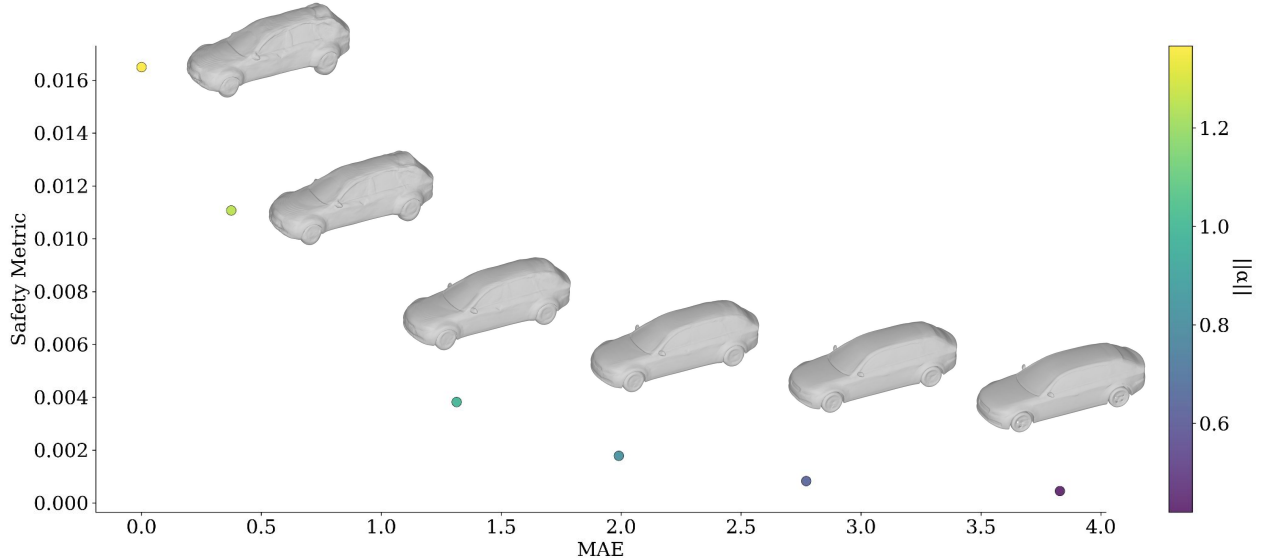


Figure 25. Trade-off between parameter error (MAE), extrapolation ($\|\alpha\|$), and geometric safety. The leftmost point illustrates conflicting parameter constraints: despite a low MAE, the high $\|\alpha\|$ and safety metric indicate that the solution lies outside the valid linear region, leading to distorted geometries after decoding.

Q. Quantifying LAMP’s High-Dimensional Extrapolation Volume

To quantify how LAMP expands the reachable design space under controlled extrapolation, we perform the analysis on a subset of 23 parameters. Each selected parameter is scaled to lie in $[-1, 1]$, and extrapolation corresponds to enlarging this 23-dimensional hypercube to

$$[-1 - f, 1 + f]^{23},$$

which has volume $(1 + f)^{23}$ relative to the original domain.

Monte Carlo Approximation

For each extrapolation factor f , we uniformly sample 10,000 random parameter vectors in the expanded domain $[-1 - f, 1 + f]^{23}$. For each sample, we solve for mixture coefficients using LAMP and generate the corresponding mesh. We then evaluate the linearity-mismatch safety metric to determine whether the geometry is valid.

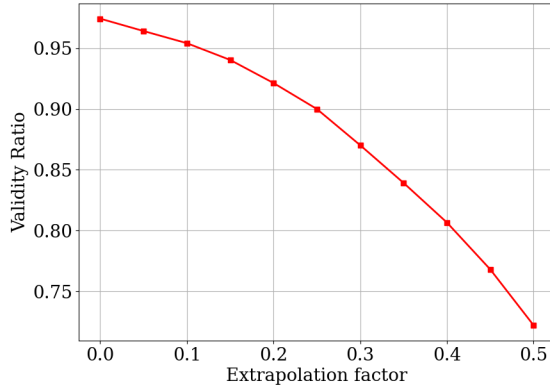
Let $\text{Valid}(f)$ denote the fraction of samples whose meshes pass the safety threshold. The **extrapolation volume ratio** for factor f is then estimated as

$$\text{VolRatio}(f) = (1 + f)^{23} \times \text{Valid}(f),$$

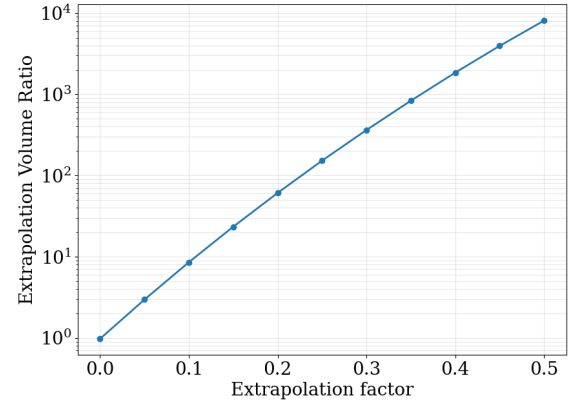
providing an aggregate measure of how much of the enlarged parameter space remains safely reachable.

Results

LAMP provides reliable parameter control even under extrapolation. Using only 100 normalized exemplar shapes, LAMP already achieves high-accuracy mixing and maintains strong local linearity. As shown in Fig. 26b, extrapolating each parameter by just 50% expands the reachable design space by a factor of $\sim 8000\times$ while still preserving a high validity ratio under the safety metric. This demonstrates that modest extrapolation combined with sufficient coverage of the parameter space enables both accurate control and large-scale geometry exploration.



(a) Validity ratio vs. extrapolation factor.



(b) Extrapolation volume ratio.

Figure 26. (a) Validity ratio under increasing extrapolation factor f . (b) Corresponding extrapolation-volume ratio in the 23-D normalized parameter space. LAMP achieves accurate control with only 100 samples and, with a 50% extrapolation ($f = 0.5$), enables exploration of a design space approximately **8000 $\times$ larger** than the dataset volume.

R. Mixing Behavior on Geometries with Topological Features

To assess whether LAMP can reliably handle shapes that include topological features such as holes, we conducted a controlled study on a simple but representative part: a rectangular plate with a circular through-hole. The geometry is parameterized by six parameters: plate *length*, *width*, and *thickness*, and hole *radius*, and hole *x*- and *y*-positions.

We trained a set of 13 exemplar SDF decoders from a shared initialization and applied LAMP to generate new shapes by varying each parameter independently. For every parameter, we performed a sweep that extends beyond the training range by $\pm 100\%$ of the original dataset bounds.

As shown in Fig. 27, LAMP successfully produces valid, smoothly varying geometries throughout these extended sweeps. In particular, the hole feature remains stable under extrapolation, and its radius and position vary as expected even when moved well outside the dataset range. These results demonstrate that LAMP is capable of mixing weights for shapes that contain localized geometric and topological features, maintaining structural coherence even under aggressive extrapolation.

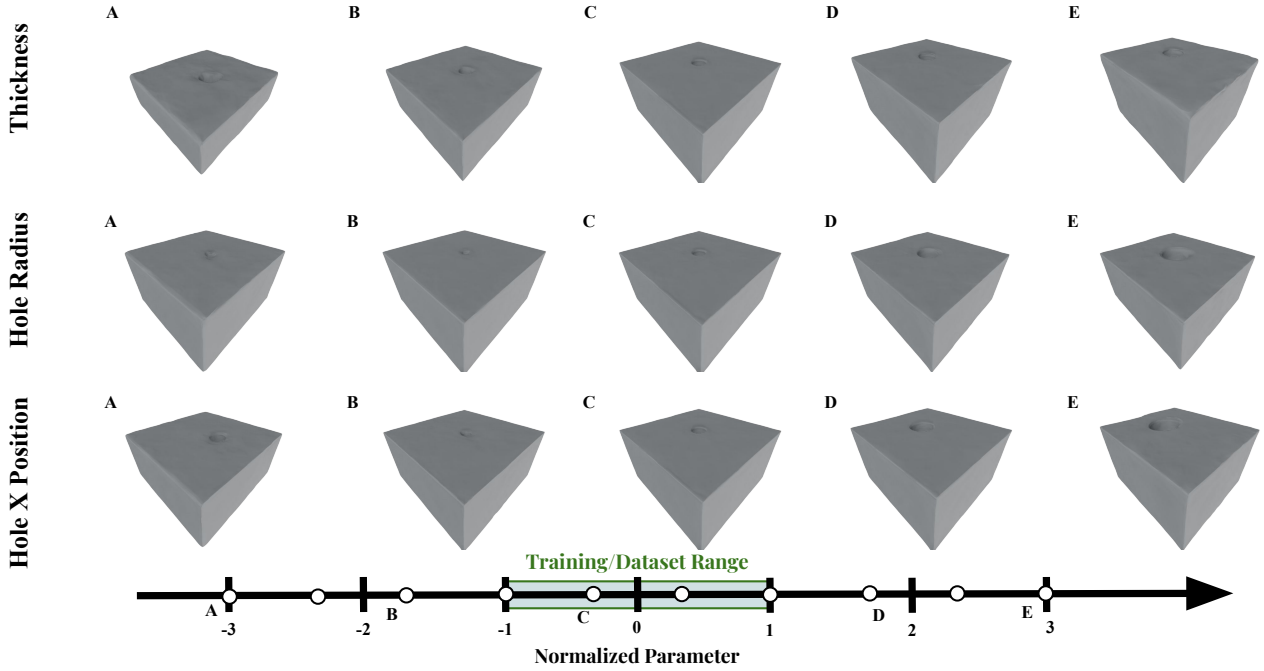


Figure 27. Parameter sweeps on a plate-with-hole geometry using LAMP. Each column (A–E) corresponds to normalized parameter values extending to $\pm 100\%$ beyond the dataset range. LAMP maintains coherent geometry and stable topological structure across all extrapolations.